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PAPER

Quantitative analysis of dynamical bifurcations in a coupled smooth and discontinuous oscillator with high-order nonlinear damping

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Abstract

The quantitative analysis of the coupled smooth and discontinuous (SD) oscillator remains challenging due to its two irrational nonlinear terms, and few studies have been conducted on it to date. This paper proposes a modified generalized harmonic function perturbation method to investigate the global evolution of limit cycles in a nonlinear damped coupled SD oscillator. This approach enables the derivation of an analytical relationship for the limit cycle amplitude as a function of the system parameters. Furthermore, an analytical stability criterion for the limit cycles is established based on the qualitative theory of ordinary differential equations. To validate the effectiveness of the proposed method, quantitative analyses are carried out for both single-well and triple-well coupled SD oscillators. In particular, critical values of the homo-heteroclinic bifurcation parameters are also predicted for the triple-well system. Additionally, approximate analytical solutions for the limit cycles and homo-heteroclinic orbits are obtained. Based on these results, global evolution of the limit cycles can be quantitatively analyzed. All analytical findings are compared with results from the Runge-Kutta method, confirming the feasibility and reliability of the proposed approach.

1. Introduction

Dynamical systems are used to describe how a system changes over time. For linear dynamical systems, there has been a very systematic and perfect theory [1]. However, with the development of science, technology and economy, linearization methods no longer meet the needs of solving complex engineering problems. The non-smooth or discontinuous factors existing in many engineering problems have gradually attracted people's attention. For example, Lu and Yang extends the Shilnikov criterion to a class of three-dimensional piecewise-smooth systems with multiple discontinuous boundaries, deriving existence conditions for chaos induced by N heteroclinic cycles [2]. Li and Wei *et al* propose a Melnikov-based framework for analyzing homoclinic chaos in piecewise smooth systems with impact and noise, verified by a seven-piecewise feedback control system [3]. Mehmood *et al* develop a discrete predator–prey model using a piecewise constant argument scheme to analyze guava pest management, identifying bifurcation conditions driven by neem-induced mortality and intervention frequency, with machine learning classifiers effectively validating the analytically derived stability regions [4]. Hu and Zhou analyze the stochastic bifurcation and chaotic dynamics of a multi-stable vibration energy harvester using a Melnikov-based approach, establishing chaos thresholds for various orbital interactions and validating noise-induced transitions through numerical simulation [5]. Li and Wei *et al* develop a non-smooth Melnikov method for hybrid piecewise-systems with unilateral rigid constraints and noise excitation, deriving chaotic criteria and demonstrating application to a mechanical system with impacts and dry friction [6]. In recent years, the discontinuous dynamical system has become a research hotspot. Cao *et al* proposed a new smooth and discontinuous archetypal oscillator (SD oscillator for short) in reference [7]. For the undamped SD oscillator, the dimensionless equation can be written as:

$$\ddot{x} + \omega_0^2 x\left(1 - \frac{1}{\sqrt{x^2 + \alpha^2}}\right) = 0. \tag{1}$$

This system is discontinuous when the smoothing parameter $\alpha = 0$, otherwise it is smooth. This equation facilitates the investigation of the dynamic transition between smooth and non-smooth regimes. There have been many studies on the dynamics associated with SD oscillators under free and forced vibration. For example, references [7, 8] have studied in detail for unperturbed SD oscillators. For forced systems with viscous damping and external harmonic excitation, the global dynamics has been studied in terms of periodic solutions [9–11], various bifurcations [12–14] and chaos [15–19]. Self-excited vibrations due to some special damping have been studied mainly in [20–23].

To examine how simple mechanical components and non-smooth nonlinearities interact in complex ways within practical engineering scenarios, Han *et al* introduced an extended version of the SD oscillator, termed the coupled SD oscillator [24]. This system takes the form of two SD oscillators coupled rigidly. The dimensionless form of the undamped coupled SD oscillator can be expressed as:

$$\ddot{x} + (x + \alpha)\left(1 - \frac{1}{\sqrt{(x + \alpha)^2 + \beta^2}}\right) + (x - \alpha)\left(1 - \frac{1}{\sqrt{(x - \alpha)^2 + \beta^2}}\right) = 0. \tag{2}$$

This system exhibits strong nonlinearity, smoothness and discontinuity. With the change of parameters α and β, the oscillator shows the characteristics of monostable, bistable and tristable. The relevant Hamiltonian for system (2) has the following form

$$H(x, y) = \frac{1}{2}y^2 + x^2 - \sqrt{(x + \alpha)^2 + \beta^2} - \sqrt{(x - \alpha)^2 + \beta^2}. \tag{3}$$

Han *et al* [24] plotted the bifurcation and the corresponding phase portraits of the oscillator (2) using the Hamiltonian function $H(x, y)$. Their study of the equilibrium stability and bifurcation problems in oscillator (2) is based on a discussion of the phase portraits. The dependence between the bifurcation sets and system parameters are constructed to illustrate the transition of potential well. Later, by using a fifth-order Taylor expansion to approximate nonlinear irrational restoring forces, Cao *et al* modeled the normal form near the catastrophe point [25]. The global bifurcation near the catastrophe point in parameter space (α, β) was studied in detail. Furthermore, directly expressing the irrational restoring force in system (2) as a polynomial using a Taylor series has the drawback of slow convergence. In order to accelerate the convergence, Lay *et al.* proposed to use rational approximation of the nonlinear restoring force function to achieve convergence truncation [26]. Then combined with harmonic balance method and Newton's method to obtain an analytical solution. The results show that low-order analytical procedures can produce highly accurate and precise solutions that are difficult to obtain with analytical expressions.

When the coupled SD oscillator is perturbed by viscous damping and external harmonic excitation, the dimensionless forced system is expressed as:

$$\ddot{x} + 2n\dot{x} + (x + \alpha)\left(1 - \frac{1}{\sqrt{(x + \alpha)^2 + \beta^2}}\right) + (x - \alpha)\left(1 - \frac{1}{\sqrt{(x - \alpha)^2 + \beta^2}}\right) = F\cos\omega\tau. \tag{4}$$

For the above oscillator (4), reference [24] used Melnikov method to obtain an analytic criterion for the chaotic thresholds. In reference [25], the nonlinear damping with general van der Pol type $(\xi + \eta x^2 + \zeta x^4)\dot{x}$ is discussed. The codimension-three bifurcation in this condition was investigated. Zhou *et al* conducted a series of studies on system (4). Firstly, for parameters $\alpha = 0.3$ and $\beta = 0.6$, Zhou *et al* investigated a bistable vibration isolation system by constructing a piecewise linearized approximation. Their analysis, based on non-smooth Melnikov theory, derived thresholds for homoclinic chaos and subharmonic resonance as functions of the damping coefficient, excitation frequency, and amplitude [27]. This study also identified infinite subharmonic bifurcations to chaos from odd-order subharmonic orbits and the coexistence of chaotic and subharmonic attractors. Secondly, for the parameter set $\alpha = 0.45$ and $\beta = 0.32$, the same authors studied coupled SD oscillators with a tristable potential [28]. For this system, they established a chaos threshold and proposed a stochastic non-smooth Melnikov method to analyze homoclinic-heteroclinic chaos under Gaussian colored noise [29]. It is worth noting that in the discontinuous limit case $\beta = 0$, equation (2) becomes the discontinuous oscillator:

$$\ddot{x} + 2x - sign(x + \alpha) - sign(x - \alpha) = 0. \tag{5}$$

Han *et al* [30] studied a coupled SD oscillator via piecewise linear approximation. They assumed that the system is perturbed by a viscous damping and an external harmonic excitation. The analytic chaos criterion for the breaking of the existing singular closed orbits of this perturbed system is successfully detected using the

Melnikov method. The complex dynamics during the transition from a single-well to a triple-well system exhibit subharmonic and chaotic solutions, along with the coexistence of multiple attractors. Chen *et al* [31] studied the limit case of the coupled SD oscillator, focusing on the scenario with Van der Pol damping. Qualitative properties of the system were obtained for all equilibria, including the infinite equilibrium point. The necessary and sufficient conditions for the existence of limit cycles and grazing cycles are established, drawing on the qualitative theory applicable to both smooth and non-smooth systems. Specifically, the study proves that the number and type of limit cycles in this oscillator are parameter-dependent, with at most two large, two small, one large double limit cycle, and three categories of grazing cycles possible. A comprehensive bifurcation analysis and a complete set of global phase portraits are detailed as well. The complex dynamics exhibited by the coupled SD oscillator—such as multi-stability, bifurcations, and sensitivity to damping—are not merely of theoretical interest. They represent a central class of problems in the design and optimization of modern micro-electro-mechanical systems (MEMS) and energy harvesters, where nonlinearities are increasingly leveraged to achieve high performance. A wide range of strategies have been developed to manage these nonlinear behaviors in resonant structures. For instance, in electrostatically actuated micro-beams, parametric instabilities and saddle-node bifurcations define the operational stability boundaries, where the role of viscous damping is critical [32]. In advanced MEMS sensors, deliberate exploitation of 1:3 internal resonance and the resulting torus or cyclic-fold bifurcations has been shown to provide novel switching mechanisms and enhanced sensitivity to external perturbations [33]. Furthermore, the ability to control dissipation mechanisms, such as thermo-elastic damping (TED), is fundamental to achieving high-quality factors in micro-resonators, with material grading and piezoelectric tuning offering promising avenues [34]. In the context of energy harvesting, the strategic design of internal resonance (e.g., 1:1 resonance) in vortex-induced vibration harvesters has demonstrated the potential to dramatically increase power output by an order of magnitude [35].

These diverse engineering applications share a common thread: they all demand a robust and accurate analytical framework capable of capturing the intricate interplay between nonlinear stiffness, high-order damping, and multi-modal interactions. The present work contributes precisely to this need. Thus, the MGHFP method is extended to the following coupled SD oscillator with high-order nonlinear damping.

$$\begin{aligned}&\ddot{x} + (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2 + \beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2 + \beta^2}}\right)\\ &= \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}. \end{aligned} \tag{6}$$

The motivation of this paper is to carry out analytical and quantitative research of dynamic bifurcations for oscillator (6). To this end, the analytical relationship between the amplitude of the steady-state periodic orbit and the system parameters needs to be derived. Meanwhile, the corresponding analytical criterion for the stability of these orbits must also be established. These results will facilitate an analytical and quantitative discussion regarding the existence, number, amplitude, and stability of each limit cycle. Additionally, it is essential to determine the analytical approximate solutions for the limit cycles as well as the homoclinic and heteroclinic orbits. However, the presence of two irrational terms in oscillator (6) renders many classical quantitative methods—such as the elliptic function L-P method [36], the elliptic perturbation method [37], and the generalized harmonic function perturbation (GHFP) method [38, 39]—incapable of accomplishing these tasks. To address this limitation, Li and Hou *et al* introduced a modified generalized harmonic function perturbation (MGHFP) method, achieved through the introduction of the composite Simpson's integration formula into the classical GHFP framework [40]. Based on this method, the dynamical bifurcations of generalized mixed Rayleigh–Liénard oscillator with cubic and quintic nonlinearities was studied analytically. Subsequently, the MGHFP method was used to analyze the generalized Duffing–Harmonic–Rayleigh–Liénard oscillator [41] and the smooth and discontinuous oscillator with quartic nonlinear damping [42]. Building on these foundations, this paper further investigates the dynamic bifurcations in oscillator (6) analytically and quantitatively by employing the MGHFP method. The analytical expressions derived from this approach quantify how the amplitude A depends on the system parameters. These expressions thereby enable the analytical and quantitative prediction of the full life period of each limit cycle, from its generation to its termination. In addition to predicting limit cycle evolution, these expressions allow for the determination of the critical values associated with homoclinic-heteroclinic bifurcations. Furthermore, the MGHFP method is employed to obtain the analytical solutions of the limit cycles and homoclinic-heteroclinic orbits of the oscillator (6). To verify the reliability and accuracy, all presented results are validated against those obtained by the Runge-Kutta method. Compared with the prior studies, the present paper extends the framework of MGHFP method to a more challenging work. First, the presence of two coupled irrational nonlinearities significantly increases the complexity of the analysis, particularly for asymmetric limit cycles. Second, the presence of quartic damping raises concerns about whether higher-order coefficients might overwhelm the perturbation parameter ε. We address this through rigorous error scaling analysis. Third, the coupled SD oscillator could exhibit triple-well potential. In this condition, homoclinic and heteroclinic orbits coexist on the same saddle point. This places higher demands on the accurate prediction of homoclinic and heteroclinic bifurcation parameters.

In addition, the mathematical model investigated in this study is characterized by irrational nonlinear stiffness, high-order nonlinear damping, and multi-stable dynamics. These properties suggest two potential applications for the system. First, the combination of high-order nonlinear damping and irrational stiffness arises naturally in vibration isolators that leverage nonlinear damping mechanisms. Recent research has focused extensively on quasi-zero stiffness (QZS) isolators incorporating nonlinear electromagnetic damping to enhance low-frequency vibration attenuation. For instance, Yuan and Jin *et al* demonstrated that integrating electromagnetic transducers into QZS isolators serves a dual purpose: mitigating nonlinear effects while simultaneously enabling energy harvesting [43]. Their work highlights how nonlinear damping characteristics profoundly shape the system's amplitude-frequency response. Similarly, Yu and Tian *et al* showed that tunable QZS isolators featuring quadratic and cubic electromagnetic damping can effectively suppress undesirable nonlinear jumps and reduce peak transmissibility in the resonance region [44]. Second, the model finds direct relevance in modern MEMS, where nonlinear damping and multi-stable dynamics are increasingly being exploited for performance enhancement. Liu and Xu *et al* combined experimental and theoretical approaches to reveal that amplitude deflection in parametrically excited MEMS resonators arises from the coupled effects of high-order nonlinearity and nonlinear damping [45]. Complementing this, Lu and Yu *et al* analyzed a MEMS model incorporating squeeze film damping and rigorously proved the existence of saddle-node bifurcations of periodic solutions. Collectively, these studies underscore the critical roles played by nonlinear damping and bifurcation-induced multi-stability in the advancement of MEMS technology [46].

The remainder of this paper is organized as follows. Section 2 presents the analytical solution for the undamped coupled SD oscillator using a nonlinear time transformation. Section 3 develops the solutions for the nonlinearly damped coupled SD oscillator via the MGHFP method. In section 4, analytical expressions are derived for the amplitude A in terms of the system parameters. Subsequently, section 5 is dedicated to numerical examples that validate the proposed method and illustrate its operational details and efficiency. Finally, a summary of conclusions is provided in section 6.

2. Free oscillation solutions of coupled SD oscillator

Let the undamped coupled SD oscillator has the following form:

$$\ddot{x} + (x + \alpha)\left(1 - \frac{1}{\sqrt{(x + \alpha)^2 + \beta^2}}\right) + (x - \alpha)\left(1 - \frac{1}{\sqrt{(x - \alpha)^2 + \beta^2}}\right) = 0, \tag{7}$$

where α and $\beta \in \mathbb{R}$. Consider its analytical solution with the initial condition:

$$x(0) = A, \dot{x}(0) = 0. \tag{8}$$

To proceed, we define a nonlinear time transformation $\varphi(t)$ given by:

$$\frac{d\varphi}{dt} = \Phi(\varphi), \Phi(\varphi + \pi) = \Phi(\varphi). \tag{9}$$

The following result is obtained by substituting equation (9) into equation (7):

$$\Phi\frac{d}{d\varphi}(\Phi x') + g(x) = 0, \tag{10}$$

where $x' = \frac{dx}{d\varphi}$, $g(x) = (x + \alpha)(1 - \frac{1}{\sqrt{(x+\alpha)^2 + \beta^2}}) + (x - \alpha)(1 - \frac{1}{\sqrt{(x-\alpha)^2 + \beta^2}})$. We establish the solution of equation (10) as

$$x(t) = a\cos^2\varphi(t) + b, \tag{11}$$

in which $a, b \in \mathbb{R}$. Multiplying through equation (10) by

$$x'(\varphi) = -2a\cos\varphi(t)\sin\varphi(t), \tag{12}$$

and we then integrate with respect to φ individually, we can get

$$\frac{1}{2}(\Phi x')^2 = -\int_{x(0)}^{x(\varphi)} g(u)du. \tag{13}$$

Let $\varphi = \pi/2$ into equation (13), we have

$$\int_{a+b}^{b} g(u)du = 0, \tag{14}$$

where $a + b = x(0) = A$. The parameters a and b are then determined directly from equation (14). From equation (13) we gain

$$\Phi(\varphi) = \sqrt{\frac{2\int_{a\cos^2\varphi+b}^{a+b} g(x)dx}{(x'_0(\varphi))^2}}, \tag{15}$$

where $\Phi(0) = \sqrt{|g(a+b)/a|}$ and $\Phi(\pi/2) = \sqrt{|g(b)/a|}$.

And Φ can be extended to a Fourier series as follows:

$$\Phi(\varphi) = \sqrt{\frac{2\int_{a\cos^2\varphi+b}^{a+b} g(x)dx}{(x'_0(\varphi))^2}} = \sum_{i=0}^{m}(p_{2i}\cos i\varphi + q_{2i}\sin i\varphi). \tag{16}$$

From the above produces, we can find the periodic solutions of the oscillator (7).

Next, we discuss its homoclinic and heteroclinic solutions. First, we consider the homoclinic solution. Its expression is also described by equation (11). Let the saddle point of the oscillator (7) is $H(h, 0)$. Then the initial value A_h of the corresponding homoclinic orbit is obtained from :

$$\int_{h}^{A_h} (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right)dx = 0. \tag{17}$$

In the limit of infinite time, a homoclinic orbit eventually approaches a saddle point. Thus, we have

$$x(\varphi_{\pm\infty}) = h, \tag{18}$$

where $\varphi_{\pm\infty} = \lim_{t\to\infty}\varphi(t)$. The integration of equation (9) gives

$$t = \int_{\varphi_0}^{\varphi} \frac{d\theta}{\Phi(\theta)}, \tag{19}$$

where φ_0 represents $\varphi(0)$. While equation (19) indicates a finite t for periodic solutions, a homoclinic solution requires an unbounded time domain. Consequently, $\Phi(\varphi)$ must satisfy the following condition:

$$\Phi(\varphi_{\pm\infty}) = 0. \tag{20}$$

It follows from equations (9) and (11) that the homoclinic orbits attain the saddle point at $x(\pm\pi/2)$, namely,

$$\varphi_{\pm\infty} = \pm\frac{\pi}{2}, \tag{21}$$

thus we obtain

$$\lim_{t\to\pm\infty} x(\varphi(t)) = x(\varphi_{\pm\infty}) = x(\frac{\pi}{2}) = b = h. \tag{22}$$

It follows from equation (15) that

$$\Phi\left(\pm\frac{\pi}{2}\right) = 0. \tag{23}$$

Given that the equation fulfills condition (20), the values of a and b can be obtained for using equations (14) and (22). The form of $\Phi(\varphi)$ is also specified by equation (15). These results together yield the homoclinic solution for oscillator (7).

Subsequently, we use the same method to investigate the heteroclinic solution of oscillator (7). We assume that the two saddle points connected by the heteroclinic orbits are $H_1(h_1, 0)$ and $H_2(h_2, 0)$. Consequently, the heteroclinic solutions must satisfy the following condition.

$$x(\varphi_{-\infty}) = h_1,\ x(\varphi_{+\infty}) = h_2. \tag{24}$$

It follows from the nonlinear time transformation (9) and solution (11) that the heteroclinic orbits attain the saddle points at positions $x(0)$ and $x(\pi/2)$, namely

$$\varphi_{-\infty} = 0,\ \varphi_{+\infty} = \frac{\pi}{2}, \tag{25}$$

then we can gain

$$\begin{cases} \lim\limits_{t\to-\infty} x(\varphi(t)) = x(\varphi_{-\infty}) = x(0) = a + b = h_1, \\ \lim\limits_{t\to+\infty} x(\varphi(t)) = x(\varphi_{+\infty}) = x(\frac{\pi}{2}) = b = h_2. \end{cases} \tag{26}$$

From equation (19), we know that the heteroclinic solution also requires an unbounded time domain. Consequently, the function $\Phi(\varphi)$ must therefore satisfy the same condition specified in equation (20). Furthermore, from an analysis of equations (11), (26) and (15), it can be inferred that

$$\Phi(\varphi_{-\infty}) = \Phi(0) = 0. \tag{27}$$

The value of h_i can be determined form equation (7), and the values of a and b can be obtained from equations (11) and (26). And $\Phi(\varphi)$ is calculated from equation (15). Finally, the heteroclinic solution is acquired.

This section mainly investigate the solutions of the undamped coupled SD oscillator. The subsequent section will present the MGHFP method, developed to analyze the coupled SD oscillator with high order non-linear damped.

3. Solutions of nonlinear damped coupled SD Oscillator with the modified generalized harmonic function perturbation method

We now direct our attention to the following coupled SD oscillator with nonlinear damping.

$$\ddot{x} + (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right) = \varepsilon f(\mu_c, x, \dot{x}). \tag{28}$$

The system involves a small positive parameter ε and a control parameter μ_c. Adopting the assumption from references [38, 39], we express the solution of oscillator (28) in the form:

$$x(t) = a\cos^2\varphi(t) + b, \tag{29}$$

where $b \in \mathbb{R}$. We expand a and Φ in powers of ε:

$$a = a_0 + \varepsilon a_1 + \varepsilon^2 a_2 + \cdots, \tag{30}$$

$$\frac{d\varphi}{dt} = \Phi(\varphi) = \Phi_0 + \varepsilon\Phi_1 + \varepsilon^2\Phi_2 + \cdots, \tag{31}$$

where Φ_0 follows from equation (15), allowing the solution (29) to be expressed as

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \cdots, \tag{32}$$

where

$$\begin{cases} x_0 = a_0\cos^2\varphi(t) + b, \\ x_n = a_n\cos^2\varphi(t)(n = 1, 2, \ldots). \end{cases} \tag{33}$$

Take equations (31) and (32) into equation (28), we have

$$\varepsilon^0\!: \Phi_0\frac{d}{d\varphi}(\Phi_0 x_0') + B_2\left(1 - \frac{1}{\sqrt{B_4}}\right) + B_1\left(1 - \frac{1}{\sqrt{B_3}}\right) = 0, \tag{34}$$

$$\begin{aligned} \varepsilon^1\!: \quad & \Phi_0\frac{d}{d\varphi}(\Phi_1 x_0') + \Phi_1\frac{d}{d\varphi}(\Phi_0 x_0') + \Phi_0\frac{d}{d\varphi}(\Phi_0 x_1') + \\ & \left(2 + \frac{B_1^2}{B_3^{3/2}} - \frac{1}{\sqrt{B_3}} + \frac{B_2^2}{B_4^{3/2}} - \frac{1}{B_4}\right)x_1 = f(\mu, x_0, \Phi_0 x_0'), \end{aligned} \tag{35}$$

$$\begin{aligned} \varepsilon^2\!: \; & \Phi_0\frac{d}{d\varphi}(\Phi_2 x_0') + \Phi_2\frac{d}{d\varphi}(\Phi_0 x_0') + \Phi_0\frac{d}{d\varphi}(\Phi_0 x_2') + \Phi_1\frac{d}{d\varphi}(\Phi_0 x_1') + \Phi_0\frac{d}{d\varphi}(\Phi_1 x_1') \\ & + \Phi_1\frac{d}{d\varphi}(\Phi_1 x_0') + \left(2 + \frac{B_1^2}{B_3^{3/2}} - \frac{1}{\sqrt{B_3}} + \frac{B_2^2}{B_4^{3/2}} - \frac{1}{B_4}\right)x_2 + 3x_1^2\left(-\frac{B_1^3}{B_3^{5/2}} + \frac{B_1}{B_3^{3/2}}\right. \\ & \left. - \frac{B_2^3}{B_4^{5/2}} + \frac{B_2}{B_4^{3/2}}\right) = f_x'(\mu, x_0, \Phi_0 x_0')x_1 + f_{\dot{x}}'(\mu, x_0, \Phi_0 x_0')(\Phi_0 x_1' + \Phi_1 x_0'), \end{aligned} \tag{36}$$

where $B_1 = x_0 - \alpha$, $B_2 = x_0 + \alpha$, , $B_4 = B_2^2 + \beta^2$, $x' = dx/d\varphi$ and $f_{\dot{x}}' = \partial f/\partial\dot{x}$. The solution to equation (34) was derived earlier, as discussed in section 2. Next, we present the steps to determine the solution of equation (35). It follows from equation (33) that

$$x_0 = a_0 \cos^2 \varphi(t) + b, \tag{37}$$

where

$$\int_{a_0+b}^{b} (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right) dx = 0, \tag{38}$$

$$\Phi_0(\varphi) = \sqrt{\frac{2\int_{a_0\cos^2\varphi+b}^{a_0+b} (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right) dx}{(x_0'(\varphi))^2}}, \tag{39}$$

By performing the multiplication of equation (35) by $x_0' = -2a_0 \cos\varphi \sin\varphi$ and subsequent integration between φ_0 and φ, one can get

$$\begin{aligned} \Phi_0\Phi_1 {x'_0}^2\Big|_{\varphi_0}^{\varphi} &= \int_{\varphi_0}^{\varphi} f(\mu, x_0, \Phi_0 x'_0) x'_0 d\varphi - \frac{a_1}{2a_0}(\Phi_0 x'_0)^2 \Bigg|_{\varphi_0}^{\varphi} \\ &-x_1\left((x_0+\alpha)\left(1 - \frac{1}{\sqrt{(x_0+\alpha)^2+\beta^2}}\right) + (x_0-\alpha)\left(1 - \frac{1}{\sqrt{(x_0-\alpha)^2+\beta^2}}\right)\right)\Bigg|_{\varphi_0}^{\varphi}. \end{aligned} \tag{40}$$

With the assignments $\varphi_0 = 0$ and $\varphi = \pi$ in equation (40), we arrive at the subsequent expression.

$$\int_0^{\pi} f(\mu, x_0, \Phi_0 x'_0) x_0' d\varphi = 0. \tag{41}$$

With the assignments $\varphi_0 = 0$ and $\varphi = \pi/2$ in equation (40), we get

$$\begin{aligned} &\int_0^{\pi/2} f(\mu, x_0, \Phi_0 x'_0) x'_0 d\varphi + a_1\left((a_0+b+\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b+\alpha)^2+\beta^2}}\right)\right. \\ &\left.+(a_0+b-\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b-\alpha)^2+\beta^2}}\right)\right) = 0. \end{aligned} \tag{42}$$

In addition, letting $\varphi_0 = 0$ in equation (40), one can get

$$\begin{aligned} &\int_0^{\varphi} f(\mu, x_0, \Phi_0 x'_0) x'_0 d\varphi - \Phi_0\Phi_1 {x'_0}^2 - \frac{a_1}{2a_0}(\Phi_0 x'_0)^2 \\ &-x_1\left((x_0+\alpha)\left(1 - \frac{1}{\sqrt{(x_0+\alpha)^2+\beta^2}}\right) + (x_0-\alpha)\left(1 - \frac{1}{\sqrt{(x_0-\alpha)^2+\beta^2}}\right)\right) \\ &+a_1\left((a_0+b+\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b+\alpha)^2+\beta^2}}\right)\right. \\ &\left.+(a_0+b-\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b-\alpha)^2+\beta^2}}\right)\right) = 0. \end{aligned} \tag{43}$$

The procedure for the first-order perturbation solution of oscillator (28) involves solving for a_0, a_1, b, Φ_0, and Φ_1. The values of a_0, b, and Φ_0 are found using equations (38), (39), and (41). Following this, a_1 is determined from equation (42), which in turn allows Φ_1 to be easily obtained from equation (43). Although higher-order terms (e.g., x_2 and Φ_2) are accessible through continued perturbation, the process becomes prohibitively complex. The first-order approximation is often adequate, primarily because it maintains acceptable accuracy even when ε is moderately large.

As the parameters α and β are varied, the oscillator (28) will appear homo–heteroclinic bifurcation. The determination of the homoclinic orbits requires that the parameters a_0 and b be solved first. Similar to section 2, we assume that the saddle point of the oscillator (28) is $H(h, 0)$. Equation (17) indicates that for a homoclinic solution, $b = 0$. The value of a_0 can then be determined from

$$\int_{x_0(0)}^{h} \left((u+\alpha)\left(1 - \frac{1}{\sqrt{(u+\alpha)^2+\beta^2}}\right) + (u-\alpha)\left(1 - \frac{1}{\sqrt{(u-\alpha)^2+\beta^2}}\right)\right) du = 0, \tag{44}$$

where $x_0(0) = a_0 + b$. Following this, Φ_0 can be solved by equation (39). The subsequent step is to substitute the solution x_0 into equation (41), from which the critical value of homoclinic bifurcation is acquired. It is noted that a real solution for μ_c from (41) constitutes a necessary condition for the homoclinic bifurcation. Finally, a_1 and Φ_1 are decided from equations (42) and (43), respectively.

Next, we outline the steps for identifying the heteroclinic orbit associated with oscillator (28). Similarly, the heteroclinic orbits are assumed to connect two saddle points, $H_1(h_1, 0)$ and $H_2(h_2, 0)$. From equation (26), we know

$$\begin{cases} a_0 + b = h_1, \\ b = h_2. \end{cases} \tag{45}$$

Following the determination of Φ_0 via equation (39), the solution x_0 is substituted into the same equation to obtain the critical value of the bifurcation parameter μ for the heteroclinic orbit. A necessary condition for the heteroclinic bifurcation is that equation (36) admits a real solution for μ_c. Finally, the parameters a_1 and Φ_1 are determined from equations (42) and (43), respectively.

4. Limit cycles and Homo–Heteroclinic orbits of coupled SD oscillator with nonlinear damping

The method from section 3 is applied here to analyze the following nonlinear damped coupled SD oscillator.

$$\begin{aligned} &\ddot{x} + (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right) \\ &= \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}. \end{aligned} \tag{46}$$

It means that

$$f(\mu_c, x, \dot{x}) = (\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x} \tag{47}$$

where $\mu_c, \mu_1, \mu_2, \mu_3, \mu_4 \in \mathbb{R}$, and μ_c is chosen as the control parameter. From equation (25), we know

$$\begin{aligned} \Phi_0(\varphi) &= \sqrt{\frac{2\int_{a_0\cos^2\varphi+b}^{a_0+b} (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right)dx}{(x'_0(\varphi))^2}} \\ &= \sum_{i=0}^{m} (p_{2i}\cos 2i\varphi + q_{2i}\sin 2i\varphi), \end{aligned} \tag{48}$$

where $m \in \mathrm{N}$. Let

$$\begin{aligned} I(\varphi) &= \int f(\mu_c, x_0, \Phi_0 x'_0) x'_0 d\varphi \\ &= \int (\mu_c + \mu_1 x_0 + \mu_2 x_0^2 + \mu_3 x_0^3 + \mu_4 x_0^4)\left(\sum_{i=0}^{m}(p_{2i}\cos 2i\varphi + q_{2i}\sin 2i\varphi)\right) x_0'^2 d\varphi. \end{aligned} \tag{49}$$

Substituting equations (37) and (48) into equation (49), the latter can be rewritten as

$$I(\varphi; a_0, b)|_0^{\pi} = 0. \tag{50}$$

Solving equations (38) and (46), a_0 and b can be determined. And substituting equation (49) into (42), we get

$$a_1 = \frac{I(\varphi)|_0^{\pi/2}}{(a_0+b+\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b+\alpha)^2+\beta^2}}\right) + (a_0+b-\alpha)\left(1 - \frac{1}{\sqrt{(a_0+b-\alpha)^2+\beta^2}}\right)}. \tag{51}$$

It follows from equation (49) that

$$a_1 = 0. \tag{52}$$

Substituting equations (49) and (52) into (43), we obtain

$$\Phi_1 = \frac{I(\varphi)}{\Phi_0 {x'_0}^2} = \sum_{i=1}^{m}(k_{2i}\cos 2i\varphi + l_{2i}\sin 2i\varphi), \tag{53}$$

where $m \in \mathbb{N}$. Hence, the expression for the limit cycle of equation (49) is as follows

$$x = a_0\cos^2\varphi + b + O(\varepsilon^2), \tag{54}$$

$$\dot{x} = -2a_0[\Phi_0 + \varepsilon\Phi_1]\cos\varphi\sin\varphi + O(\varepsilon^2). \tag{55}$$

In general, the Fourier coefficients p_{2i} and q_{2i} in equation (49) cannot be obtained analytically when $\Phi_0(\varphi)$ is relatively complex. This means that it is difficult to derive an analytical relationship between the amplitude A of the limit cycle and the parametesr μ_i. The oscillator (46) can be solved only if all system parameters are given. In order to overcome this drawback, the following optimization scheme is proposed.

Let $m = 4$ in equation (48), that is, $\Phi_0(\varphi) = \sum_{i=0}^{4} p_{2i}\cos 2i\varphi$, and bring it into equation (50), one obtains the following analytical relationship between the amplitude A of the limit cycle and the parameter μ_c.

$$\mu_c = \frac{4}{\pi a_0^2(2p_0 - p_4)}(E_1\mu_1 + E_2\mu_2 + E_3\mu_3 + E_4\mu_4), \tag{56}$$

where $p_{2i}(i = 0, 1, 2, 3, 4)$ are Fourier coefficients and

$$\begin{aligned}
E_1 &= \frac{1}{16}\pi a_0^2(a_0(-4p_0 - p_2 + 2p_4 + p_6) + 4b_0(p_4 - 2p_0)),\\
E_2 &= \frac{1}{64}\pi a_0^2(-8a_0b_0(4p_0 + p_2 - 2p_4 - p_6) + a_0^2(-10p_0 - 4p_2 + 4p_4 + 4p_6 + p_8)\\
&\quad + 16b_0^2(p_4 - 2p_0)),\\
E_3 &= \frac{1}{256}\pi a_0^2(-12a_0^2b_0(10p_0 + 4p_2 - 4p_4 - 4p_6 - p_8) - 48a_0b_0^2(4p_0 + p_2 - 2p_4 - p_6)\\
&\quad + a_0^3(-28p_0 - 14p_2 + 8p_4 + 13p_6 + 6p_8) + 64b_0^3(p_4 - 2p_0)),\\
E_4 &= \frac{1}{1024}(\pi a_0^2(-16a_0^3b_0(28p_0 + 14p_2 - 8p_4 - 13p_6 - 6p_8) - 96a_0^2b_0^2(10p_0 + 4p_2 - 4p_4\\
&\quad - 4p_6 - p_8) - 256a_0b_0^3(4p_0 + p_2 - 2p_4 - p_6) + a_0^4(-84p_0 - 48p_2 + 15p_4 + 40p_6 + 26p_8)\\
&\quad + 256b_0^4(p_4 - 2p_0))).
\end{aligned}$$

The p_{2i} in equation (56) are calculated by employ the compound Simpson integration formula [47] with dividing the integral interval into eight equal parts. The expressions of p_{2i} are listed in Appendix A.

Next, from equation (38), we give the relationship between the amplitude of the limit cycle and the coefficients a_0 and b. For symmetric limit cycles, equation (38) can be rewritten in the following form

$$\int_A^{-A}(x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right)dx = 0, \tag{57}$$

where A denotes the amplitude of limit cycles, simultaneous equation (57) along with $b = -A$ and $a_0 + b = A$. Thus, we obtain

$$a_0 = 2A,\ b = -A. \tag{58}$$

For asymmetric limit cycles, from equation (38) and $a_0 + b = A$, we can get

$$a_0 = A - C,\ b = C. \tag{59}$$

Due to the presence of a dual-irrational term, the explicit solution for a_0 and b is somewhat complicated. Therefore, the expressions for C are provided in Appendix B. Once the values of α and β are assigned, a_0 and b can be expressed in terms of the amplitude A.

Remark 1. From reference [25], it can be seen that as the parameters α and β are varied, the oscillator (46) will have double well and triple well. At this time, oscillator (46) exists homo–heteroclinic bifurcation. Setting $b = h$ yields the critical value of μ_c for the homoclinic bifurcation. Similarly, setting $b = h_2$ and $a_0 + b = h_1$ yields the critical μ_c for the heteroclinic bifurcation.

In addition, the stability of a limit cycle is evaluated by introducing the following characteristic quantity.

$$h_0 = \int_0^{\pi}\frac{1}{\Phi_0}\left(\mu + \sum_{i=1}^{4}\mu_i(a_0\cos^2\varphi + b)^i\right)d\varphi, \tag{60}$$

where the values of a_0 and b have been obtained in equations (58) and (59).

Remark 2. Based on the qualitative theorem of ordinary differential equations [48], a limit cycle is unstable when its $h_0 > 0$, semi-stable when its $h_0 = 0$, and stable when its $h_0 < 0$.

Remark 3. This paper derives analytical expressions of Fourier coefficients p_{2i}. One can also calculate them by numerical integration directly. While a single direct numerical integration may be faster than a single execution of the analytical procedure, the advantage of the proposed method becomes evident in parameter studies. Once the amplitude equation (as equation (56)) is derived, scanning hundreds of parameter values reduces to simple algebraic evaluations, requiring substantially less computational time. In contrast, direct numerical integration would need to be repeated from scratch for each parameter set, incurring significant cumulative cost. Thus, the method is particularly suitable for tasks requiring extensive parametric exploration or stability analysis, where the upfront analytical effort pays off. Conversely, when the task is simply to locate limit cycle amplitudes for a fixed parameter set, direct numerical integration can be a practical alternative.

This section focuses on the methods for solving both the limit cycles and the homoclinic/heteroclinic orbits in the nonlinear damped coupled SD oscillator. Specifically, we derive the analytical relationship between the amplitude of the limit cycle and the control parameter. This relationship not only characterizes the system's periodic motion but also enables the prediction of critical homoclinic and heteroclinic bifurcation parameters. Its feasibility and accuracy will be verified through two examples in the following section.

5. Examples

This section employs two examples to investigate the limit cycles and homo-heteroclinic bifurcations in the nonlinear damped coupled SD oscillators (46). Among them, Example 1 corresponds to a single-well potential, while Example 2 addresses a triple-well potential. The present results are benchmarked against numerical solutions obtained via the Runge-Kutta method to ensured the validity.

5.1. Example for single well potential

This example consider the following couple SD oscillator with the condition of single well potential.

$$\begin{aligned}&\ddot{x} + (x+\alpha)\left(1 - \frac{1}{\sqrt{(x+\alpha)^2+\beta^2}}\right) + (x-\alpha)\left(1 - \frac{1}{\sqrt{(x-\alpha)^2+\beta^2}}\right)\\ &= \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}.\end{aligned} \tag{61}$$

First, let $\alpha = 2$, $\beta = 0.5$, and $\mu_1 = 1$, $\mu_2 = 1$, $\mu_3 = 1$, $\mu_4 = -1$. Substituting them into equations (56) and (60), we can obtain the relationship between the amplitudes A and parameters μ_i. As well as, the relationship between A and h_0 is also acquired. The curves of $\mu_c - A$ and $h_0 - A$ are plotted in figure 1. The upper plot in this figure represents the relationship between the limit cycle amplitude A and the control parameter μ_c for oscillator (61). And the lower plot represents the relationship between A and the characteristic quantity h_0. In figure 1, $\mu^{(0)} = -0.1249$, $\mu^{(1)} = -0.05$ and $\mu^{(2)} = 0.02$ are three special values of parameter μ_c. $A^{(0)}(1, -0.1249)$, $A^{(11)}(0.4749, -0.05)$, $A^{(12)}(1.331, -0.05)$ and $A^{(2)}(1.4186, 0.02)$ are coordinates of four marked points. This figure clearly depicts the evolution of the limit cycles as a function of the control parameter μ_c, demonstrating the following characteristics.

(i) When $\mu_c < \mu^{(0)}$, no limit cycle exists.

(ii) With μ_c increasing near $\mu^{(0)}$, a limit cycle emerges with amplitude $A^{(0)}$. Its characteristic quantity, shown in figure 1 to be zero, identifies it as semi-stable. Analysis of the lower plot in figure 1 shows that near $A^{(0)}$, the curve crosses the abscissa from positive to negative. This crossing behavior implies an outer-stable, inner-unstable configuration for this semi-stable limit cycle.

(iii) Further increase of μ_c within the range $\mu^{(0)} < \mu_c < 0$ causes the semi-stable limit cycle to bifurcate into two: a smaller unstable cycle ($h_0 > 0$) and a larger stable cycle ($h_0 < 0$), as evidenced by figure 1. At the critical value $\mu_c = 0$, the unstable cycle collapses to the singular point, while the stable one grows further.

(iv) When the parameter $\mu_c > 0$, only a single stable limit cycle persists, whose amplitude grows with increasing μ_c.

In addition to the above analysis, the method can be extended to calculate the approximate solution for oscillator (61). Considering the four cases of $\mu_c = -0.12$, $\mu_c = -0.05$, $\mu_c = -0.01$ and $\mu_c = 0.02$, the corresponding value of A can be obtained by using equations (56) and (58). For different values of A, the corresponding Φ_0 and Φ_1 can be obtained from equations (48) and (53). From this, the first order approximate solution of the limit cycle can be expressed as:

$$x = 2A\cos^2\varphi - A + O(\varepsilon^2) \tag{62}$$

$$\frac{d\varphi}{dt} = \Phi_0(\varphi) + \varepsilon\Phi_1(\varphi) + O(\varepsilon^2) \tag{63}$$

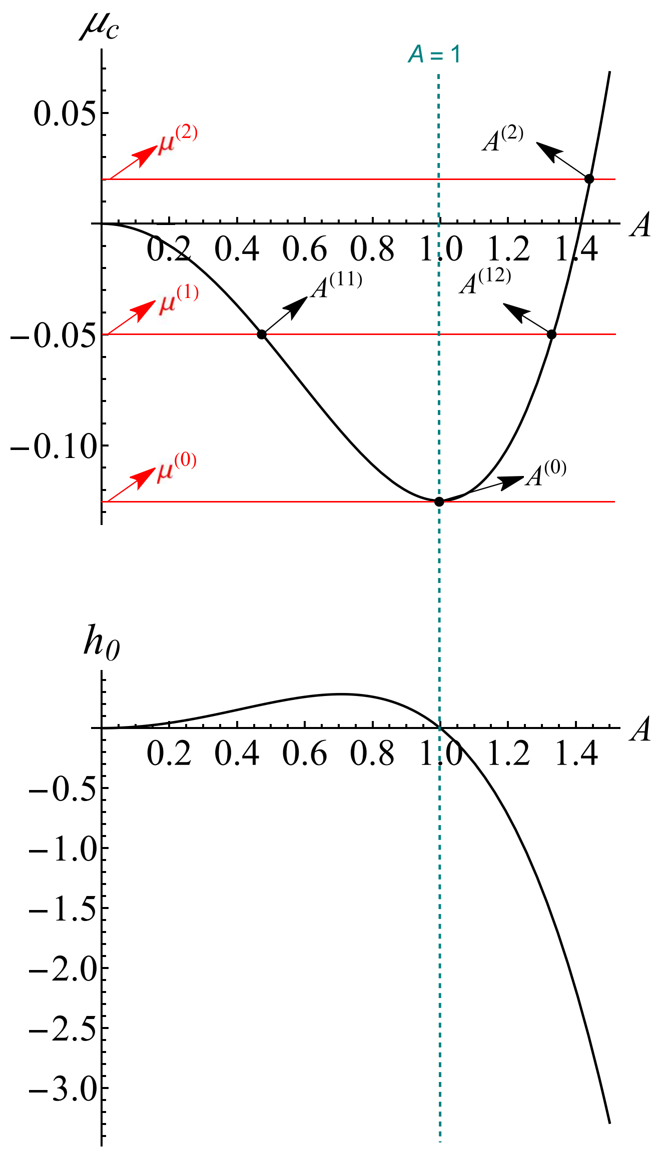


Figure 1. The relationship between the amplitude A and control parameter μ_c as well as characteristic quantity h_0 of the oscillator (61).

$$\Phi_0 = p_0 + p_2\cos(2\varphi) + p_4\cos(4\varphi) + p_6\cos(6\varphi) + p_8\cos(8\varphi) + HH \tag{64}$$

$$\Phi_1 = l_2\sin(2\varphi) + l_4\sin(4\varphi) + l_6\sin(6\varphi) + l_8 sin(8\varphi) + HH \tag{65}$$

where HH denotes the higher order harmonic terms The values of A, p_i and l_i for each μ_c are listed in table 1. Figure 2 displays the phase portraits of the solutions corresponding to $\varepsilon = 0.2$. It is not difficult to see from the diagram that the results obtained by present method are also accurate for moderately large ε when compared with Runge-Kutta method. From the image of the solutions, these results are consistent with that shown in figure 1, that is, when $\mu^{(0)} < \mu_c < 0$, the amplitude of the larger limit cycle gradually increases with the increase of μ_c, while the amplitude of the smaller limit cycle gradually decreases. As μ_c approaches 0, the smaller limit cycle progressively shrinks toward the equilibrium point (0, 0). Concurrently, the larger limit cycle monotonically increases in amplitude with μ_c. To further display the accuracy of solutions depicted in this figure, the maximum absolute error in amplitude (MAEA), the root mean square phase error over one period (RMSPE) and relative L_2-norm differences (L_2-ND) are calculated and established in table 2. The indicators in this table all illustrate that the obtained solution has high accuracy when $\varepsilon = 0.2$. Meanwhile, the influence of the parameter ε on the accuracy of the solutions is also investigated. Detailed quantitative results, including the MAEA, RMSPE, and L_2-ND for varying ε values corresponding to the solutions in figure 2(a), are presented in table 3. It evident that, if the absolute values of $\varepsilon\mu_i \leqslant 0.5 (i = 1, 2, 3, 4)$, the accuracy of the present approximate solutions it acceptable.

Furthermore, due to the presence of the high-order nonlinear damping term $\varepsilon\mu_4 x^4\dot{x}$, a crucial question arises as to whether this term could overwhelm the perturbation parameter ε and thereby compromise the

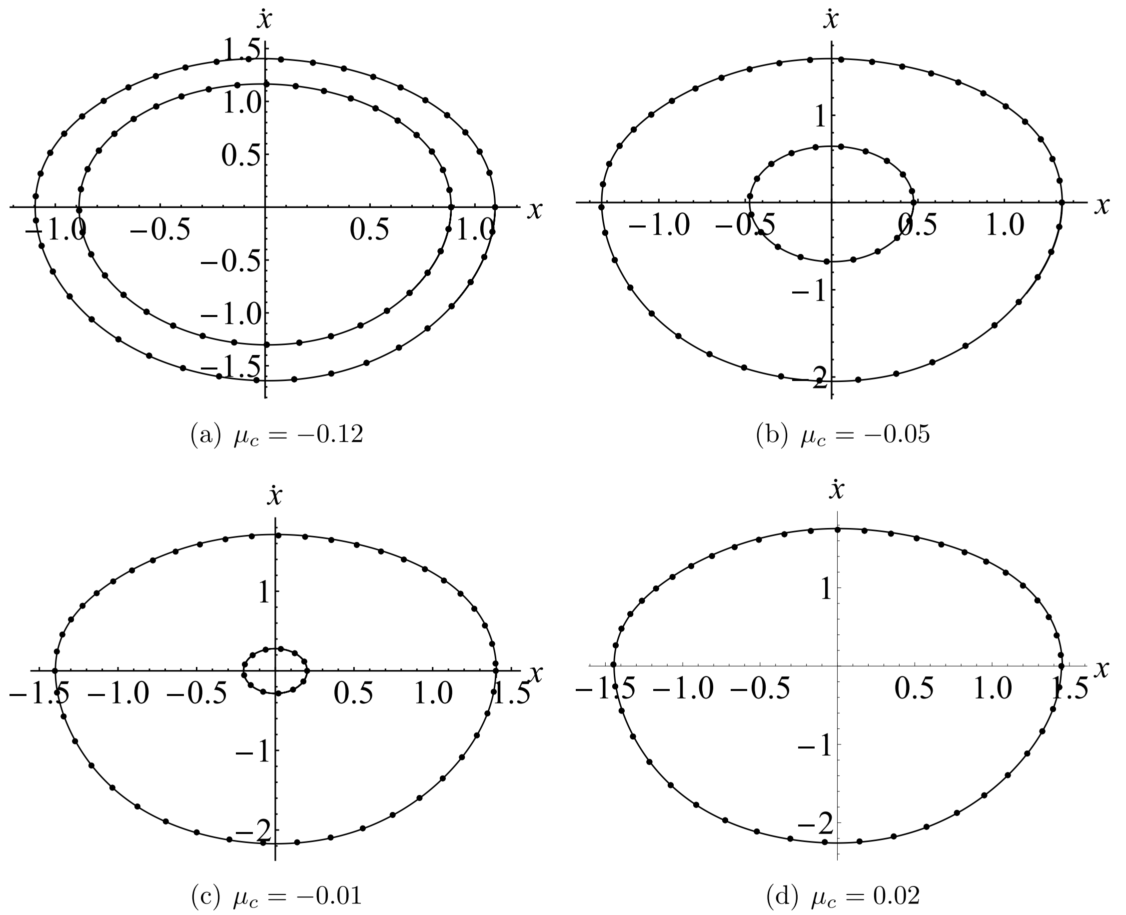


Figure 2. Limit cycles for oscillator (61) at different μ_c when $\mu_1 = 1, \mu_2 = 1, \mu_3 = 1, \mu_4 = -1$ and $\varepsilon = 0.2$. '—' denote the results via the Runge-Kutta method. '···' denote the results via the present method.

Table 1. The values of A, p_i and l_i for each μ_c in solutions (62)–(65).

μ_c	-0.12	-0.05	-0.01	0.02
A	0.8865/1.0959	0.4749/1.3331	0.2021/1.4011	1.4482
p_0	0.6936/0.6913	0.6961/0.6867	0.6968/0.6848	0.6822
p_2	0/0	0/0	0/0	0
p_4	−0.0012/−0.0023	−0.0003/−0.0046	−0.0005/−0.0056	−0.0064
p_6	0/0	0/0	0/0	0
p_8	−0.0001/−0.0002	0/−0.0046	0/−0.0006	−0.0008
l_2	0.2114/0.327	0.0889/0.4376	0.0344/0.4832	0.5168
l_4	0.0169/−0.0001	0.0114/−0.0533	0.0025/−0.0778	−0.0977
l_6	0.0175/0.0331	0.0027/0.0596	0.0002/0.092	0.0764
l_8	−0.0064/−0.0150	−0.0005/−0.0329	0/−0.0402	−0.0459

asymptotic validity of the first-order solution. To address this issue, we take the larger limit cycle depicted in figure 2(a) as an example and perform the following linear regression of its errors E_i (including MAEA, RMSPE, and L_2-ND) against $\ln \varepsilon$.

$$\ln E_i = \ln C_i + k_i \ln \varepsilon, \ \varepsilon \in [0.1, \ 0.7] \tag{66}$$

where C_i is a constant. It means that

$$E_i = \mathrm{e}^C \times \varepsilon^{k_i}, \ \varepsilon \in [0.1, \ 0.7] \tag{67}$$

where k_i is called scaling exponent. The regression coefficients are established in table 4. In perturbation analysis, the validity of a solution relies on the perturbation parameter ε being a small parameter in the system, and all approximate terms are estimated in orders of powers of ε. For a first-order perturbation solution (retaining terms up to ε^1), theoretically, the error should primarily originate from the neglected higher-order

Table 2. Error analyses of the solutions in figure 2.

Figures	Amplitudes	MAEA [a]	RMSPE [b]	L_2-ND [c]
(a)	0.89521	0.0030628	0.00219737	0.442%
	1.09507	0.0017032	0.00425054	0.398%
(b)	0.47482	0.0001218	0.00078690	0.047%
	1.33312	0.0071334	0.01226741	1.129%
(c)	0.20208	9.2×10^{-6}	0.00014307	0.007%
	1.40110	0.0109716	0.01849910	1.463%
(d)	1.44290	0.0139366	0.02219521	1.706%

[a] MAEA denotes the maximum absolute error in amplitude.
[b] RMSPE denotes the root mean square phase error over one period.
[c] L_2-ND denotes the relative L_2-norm differences.

Table 3. Error analyses with respect to ε of the solutions in figure 2(a).

Amplitudes	Indicators	$\varepsilon = 0.1$	$\varepsilon = 0.3$	$\varepsilon = 0.4$	$\varepsilon = 0.5$	$\varepsilon = 0.7$
1.09507	MAEA [a]	0.0003869	0.0041133	0.0079901	0.0140332	0.0398905
	RMSPE [b]	0.0010937	0.0094863	0.0170484	0.0275539	0.0630447
	L_2-ND [c]	0.103%	0.870%	1.531%	2.421%	5.383%
0.89521	MAEA [a]	0.0007429	0.0071037	0.0131636	0.0218212	0.0533222
	PEOP [b]	0.0008567	0.0054962	0.0092576	0.0140032	0.0272224
	L_2-ND [c]	0.108%	1.016%	1.852%	2.993%	6.690%

[a] MAEA denotes the maximum absolute error in amplitude.
[b] RMSPE denotes the root mean square phase error over one period.
[c] L_2-ND denotes the relative L_2-norm differences.

Table 4. Regression coefficients of E_i in equation (67).

E_i	C_i	k_i	R^{2} [a]
MAEA	−2.5832826	2.3360164	0.9943880
RMSPE	−2.1304141	2.0590161	0.9980955
L_2-ND	2.3136776	2.0082027	0.9986052

[a] R^2 is the coefficient of determination, measuring how well the regression model fits the data—values close to 1 indicate an excellent fit.

terms (i.e., ε^2 and beyond). Therefore, if the scaling exponent k_i is close to the theoretical value of 2, it indicates that the first-order truncation error dominates, and the contribution of the damping term $\mu_4 x^4 \dot{x}$ has been effectively averaged out without introducing additional low-order errors. The data presented in table 4 support this statement. It indicates that although the absolute value of μ_4 is equal to 1, the accuracy of the first-order perturbation solution obtained in this paper is still dominated by ε.

Finally, as shown in figure 3, for the oscillator (61), when $\varepsilon = 0.1$, we consider twenty different sets of parameters. And then substitute them into the equation (56) to obtain the relationship between parameter μ_c and amplitude A, respectively. Based on this figure, we can get the following inferences. Given other parameters, it can be seen from figure 3(a), the curves have no extreme point when $\mu_4 = 0$, $\mu_4 = 0.5$ and $\mu_4 = 1$, which implies that the oscillators have only one limit cycle. However, when $\mu_4 = -0.5$ and $\mu_4 = -1$, curves have an extreme point, which means that two limit cycles exist for the oscillators. Therefore, we can infer that when $\mu_4 \geqslant 0$, only one limit cycle exists, and two limit cycles appear when $\mu_4 < 0$. Similarly, it can be seen from figure 3(b), the curves have no extreme points when $\mu_2 = -1$, $\mu_2 = -0.5$ and $\mu_2 = 0$, which implies that the oscillators have only one limit cycle, and the curves have one extreme point when $\mu_2 = 0.5$ and $\mu_2 = 1$, which means that there are two limit cycles. Therefore, we can infer that when $\mu_2 \leqslant 0$, there is only one limit cycle exists and two limit cycles appear when $\mu_2 > 0$. In the same way, we can infer from figures 3(c) and (d) respectively. Figure 3(c) has two limit cycles when $\mu_4 > 0$ and only one limit cycle when $\mu_4 \leqslant 0$. For $\mu_2 < 0$, figure 3(d) has two limit cycles, while for $\mu_2 \geqslant 0$ there is only one limit cycle. Finally, to show the accuracy, the above curves

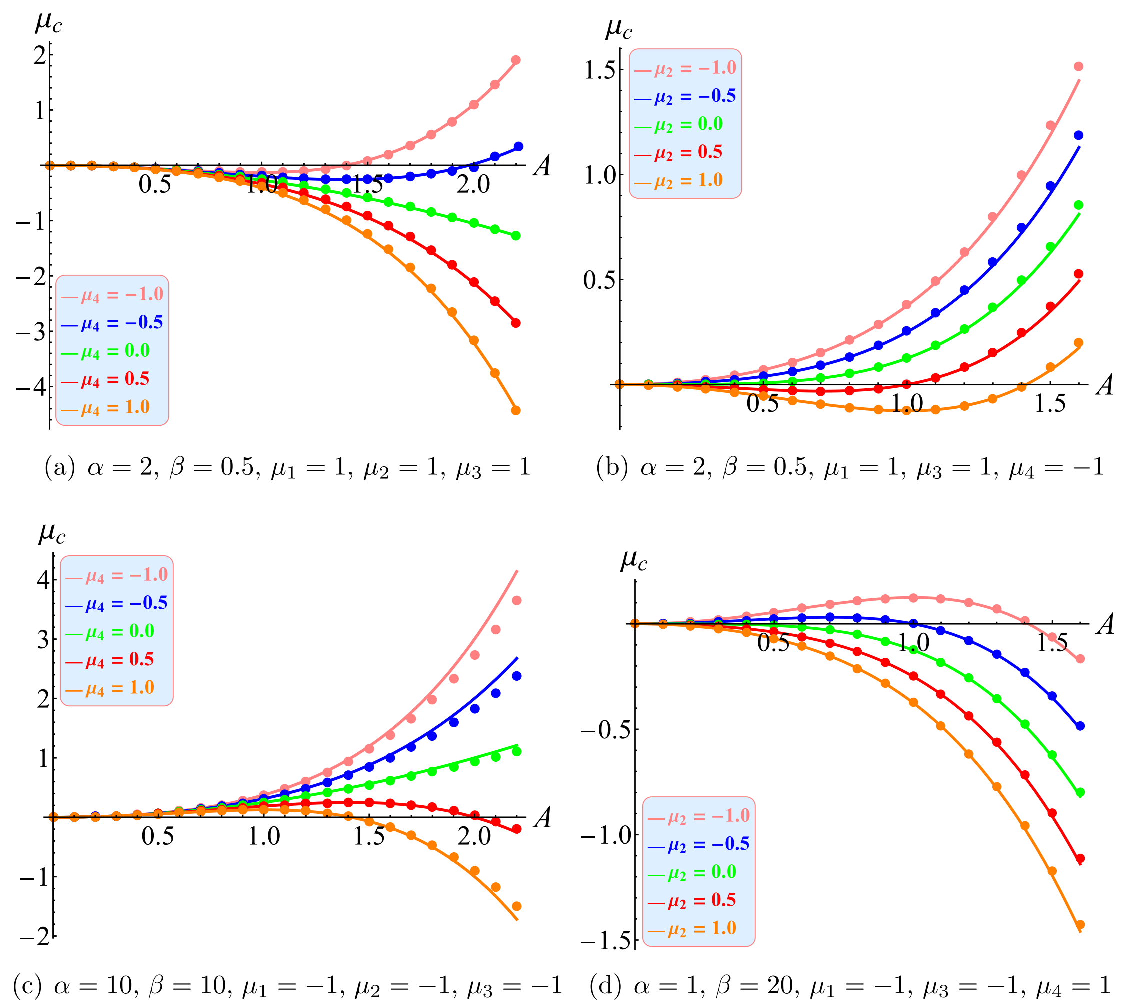


Figure 3. The relationship between the amplitude A and control parameter μ_c of the oscillator (61) with different parameters as $\varepsilon = 0.1$. '—' denote the results via the present method. '⋯' denote the results via the Runge-Kutta method.

are also drawn by numerical method when $\varepsilon = 0.1$ and shown together in figure 3. Through comparison, it can be found that the present method has high precision and the predicted results are reliable.

As we all know, the oscillator (61) contains a smoothness parameter β. When $\beta = 0$ and $\alpha \geqslant 1$, equation (61) reduces to a non-smooth single-well oscillator. To assess the accuracy of the present method under this condition, the $\mu_c - A$ curves for $\beta = 0$ are plotted in figure 4. A comparison with figure 3 reveals that the non-smooth factor influences the accuracy of the results obtained by our method. Nevertheless, if $|\varepsilon\mu_i|$ is sufficiently small and the amplitude satisfies $A \leqslant 2$, the predictions of the proposed method remain reliable.

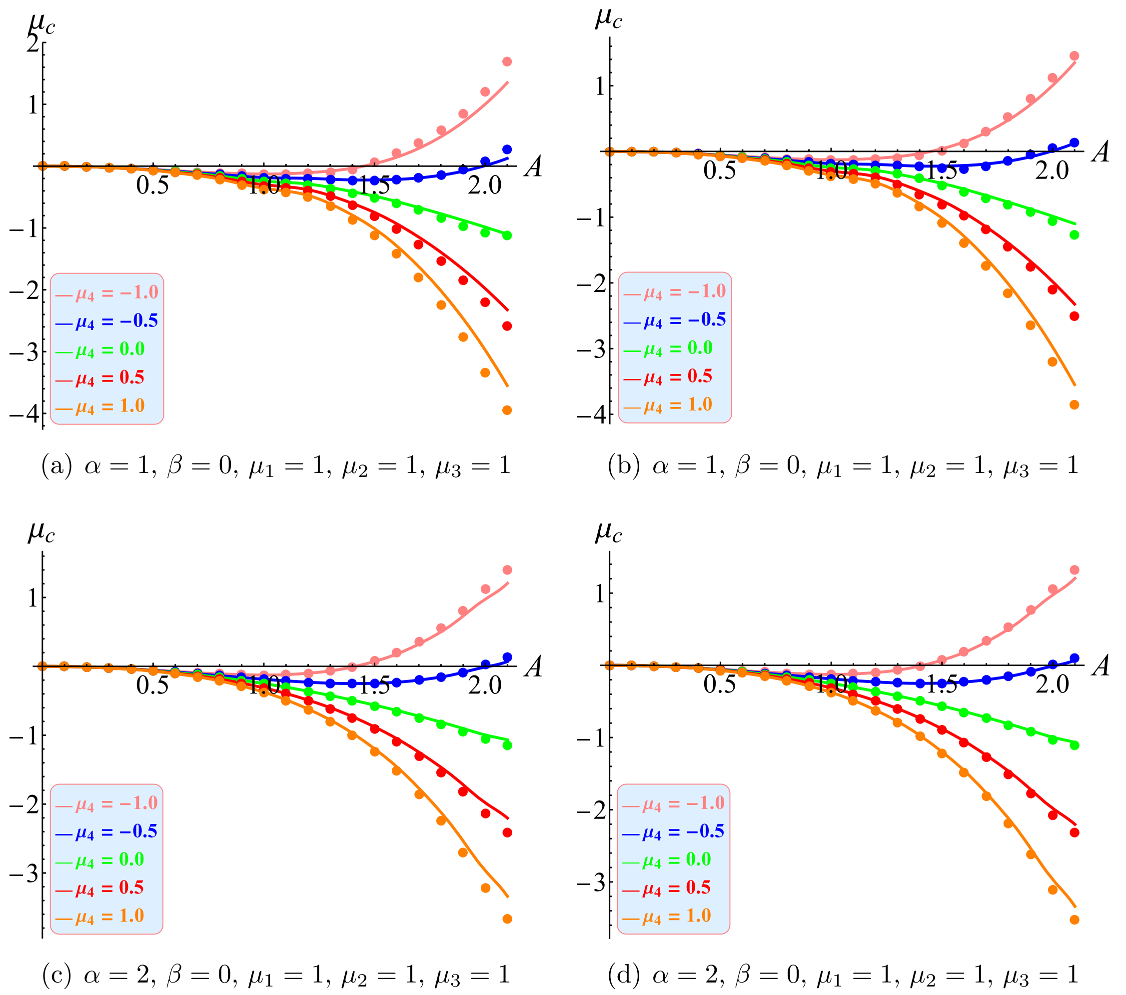
5.2. Example for triple well potential

This example consider the following couple SD oscillator with the condition of triple well potential.

$$\ddot{x} + (x + \alpha)\left(1 - \frac{1}{\sqrt{(x + \alpha)^2 + \beta^2}}\right) + (x - \alpha)\left(1 - \frac{1}{\sqrt{(x - \alpha)^2 + \beta^2}}\right)$$
$$= \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x} \tag{68}$$

The potential portrait of it is shown in figure 5. It can be seen from figure 5 that the system has five equilibrium points, in which H_1, H_3 and H_5 are the central points, H_2 and H_4 are the saddle points, and S_1 and S_2 are the equipotential points of the saddle points H_2 and H_4. Obviously, between S_1 and H_2 (S_2 and H_4), there are asymmetric periodic orbits. There are symmetric periodic orbits inside points H_2 and H_4, as well as outside the points S_1 and S_2. At the same time, saddle points H_2 and H_4 are connected with a homoclinic orbit respectively, and a heteroclinic orbit is connected between them.

Next, take $\alpha = 0.5$, $\beta = 0.1$, $\mu_1 = 2$, $\mu_2 = -0.1$, $\mu_3 = -1$, and $\mu_4 = 0.7$ as an example, then we get $H_1 = -0.988726739$, $H_2 = -0.500619604$, $H_4 = 0.988726739$ and $H_5 = 0.500619604$, $S_1 = -1.391467466$. Substituting system parameters into equations (56) and (60), we can get the relationship between the



(a) $\alpha = 1,\ \beta = 0,\ \mu_1 = 1,\ \mu_2 = 1,\ \mu_3 = 1$ (b) $\alpha = 1,\ \beta = 0,\ \mu_1 = 1,\ \mu_2 = 1,\ \mu_3 = 1$

(c) $\alpha = 2,\ \beta = 0,\ \mu_1 = 1,\ \mu_2 = 1,\ \mu_3 = 1$ (d) $\alpha = 2,\ \beta = 0,\ \mu_1 = 1,\ \mu_2 = 1,\ \mu_3 = 1$

Figure 4. The relationship between the amplitude A and control parameter μ_c of the oscillator (61) in its non-smooth case as (a)(c) $\varepsilon = 0.1$, (b)(d) $\varepsilon = 0.05$. '—' denote the results via the present method. '$\cdots$' denote the results via the Runge-Kutta method.

amplitudes A and parameters . As well as, the relationship between A and characteristic quantity is also obtained. Their graphs are shown in figure 6. In this case, the structure of solutions in phase plane is separated by its homoclinic orbit and heteroclinic orbit. Therefore, the graphs of $\mu_c - A$ and $h_0 - A$ are plotted separately when $0 < A < 0.500619604$, $0.500619604 < A < 1.391467466$ and $A > 1.391467466$. Here 0.500619604 is the initial value of homo-heteroclinic orbit.

In figure 6, $\mu_{IL}^{(0)} = 0.00177$, $\mu_{IL}^{(1)} = 0.0015$, $\mu_{IL}^{(2)} = 0.00126798$, $\mu_s^{(1)} = -1.603$, $\mu_s^{(2)} = -1.60023$, $\mu_s^{(3)} = -1.5821$, $\mu_{OL}^{(0)} = -1.606$ and $\mu_{OL}^{(1)} = -0.47242$ are special values of parameter μ_c. $A_{IL}^{(0)}(0.3785,\ 0.00177)$, $A_{IL}^{(11)}(0.2939,\ 0.0015)$, $A_{IL}^{(12)}(0.4532,\ 0.0015)$, $A_{IL}^{(21)}(0.2574,\ 0.00126798)$, $A_{IL}^{(22)}(0.5,\ 0.00126798)$, $A_s^{(01)} = (0.5809,\ -1.60509)$, $A_s^{(02)} = (1.363,\ -1.60509)$, $A_s^{(11)} = (0.5354,\ -1.603)$, $A_s^{(12)} = (0.6349,\ -1.603)$, $A_s^{(13)} = (1.3235,\ -1.603)$, $A_s^{(13)} = (1.3855,\ -1.603)$, $A_s^{(21)} = (0.6706,\ -1.60023)$, $A_s^{(22)} = (1.2935,\ -1.60023)$ and $A_{OL}^{(0)} = (2.02586,\ -1.606)$ are coordinates of fourteen marked points. The subscript IL denotes the internal large limit cycle (internal symmetric limit cycle) which encloses only one singular point (0,0) in phase plane. The subscript s denotes the small limit cycle (asymmetric limit cycle) which encloses only one singular point in phase plane. The subscript OL denotes the outer large limit cycle (outer symmetric limit cycle) which encloses all singular points in phase plane. As illustrated, this figure similarly depicts the dependence limit cycles on μ_c. In contrast to the symmetric limit cycles described in Example 1, the focus here shifts to analyzing asymmetric ones, that is the limit cycles described in figure 6(b).

(i) When $\mu_c > \mu_s^{(0)}$, no limit cycle exists.

(ii) With μ_c increasing near $\mu_s^{(0)}$, a small limit cycle emerges with amplitude $A_s^{(01)}(A_s^{(02)})$. Its characteristic quantity, shown in figure 6(b) to be zero, identifies it as semi-stable. Analysis of the lower plot in figure 6(b) shows that near $A_s^{(01)}$, the curve crosses the abscissa from negative to positive. This crossing behavior implies an outer-unstable, inner-stable configuration for this semi-stable limit cycle.

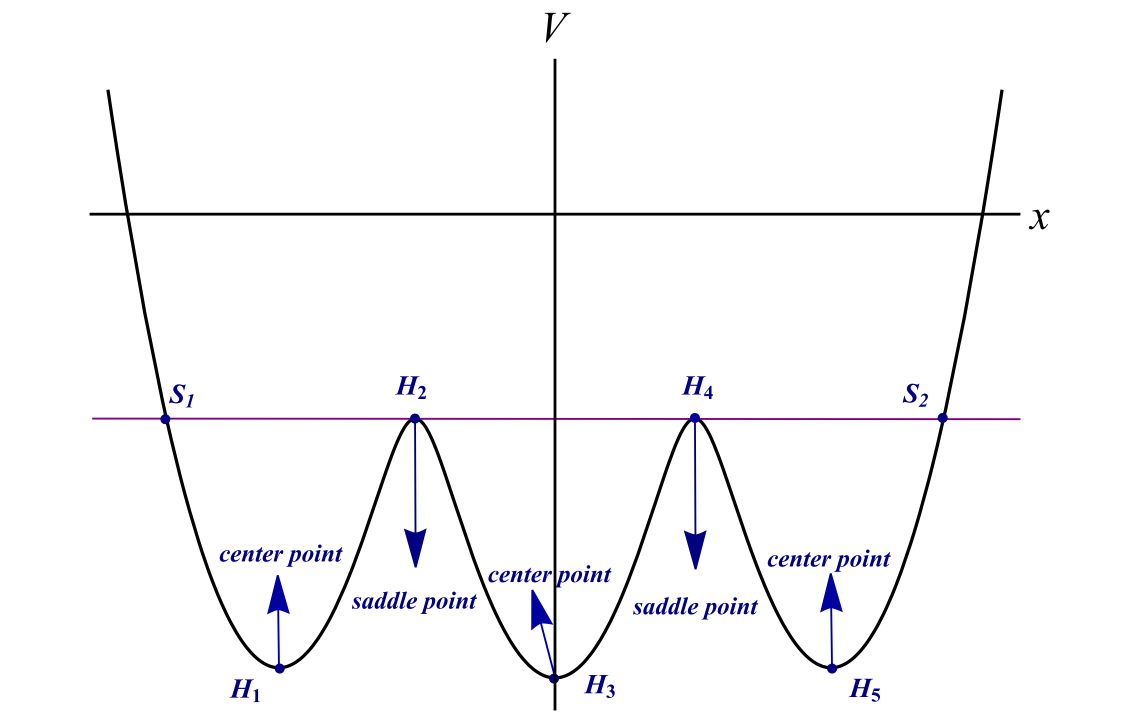


Figure 5. Potential energy diagram of the oscillator (68) in triple well potential.

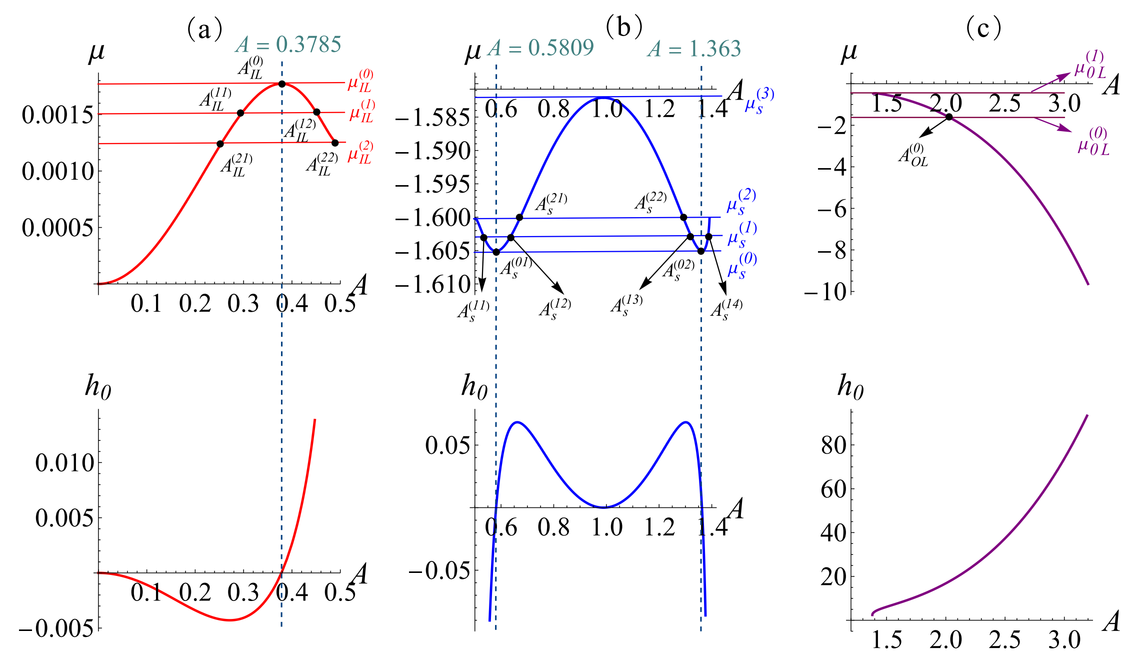


Figure 6. The relationship between the amplitude A and control parameter μ_c as well as h_0 of oscillator (68).

(iii) As parameter μ_c continues to increase, the semi-stable small limit cycle bifurcates to two small limit cycles when $\mu_s^{(0)} < \mu_c < \mu_s^{(2)}$. From figure 6(b), we can see that the characteristic quantity $h_0 > 0$ for the smaller small limit cycle and $h_0 < 0$ for the larger small limit cycle. Thus, the smaller small limit cycle is unstable and the larger small limit cycle is stable.

(iv) Especially, when $\mu_c = \mu_s^{(2)}$, the smaller small limit cycle still exists, while the larger small limit cycle converges to homoclinic orbit which means the oscillator (68) occurs the homoclinic bifurcation.

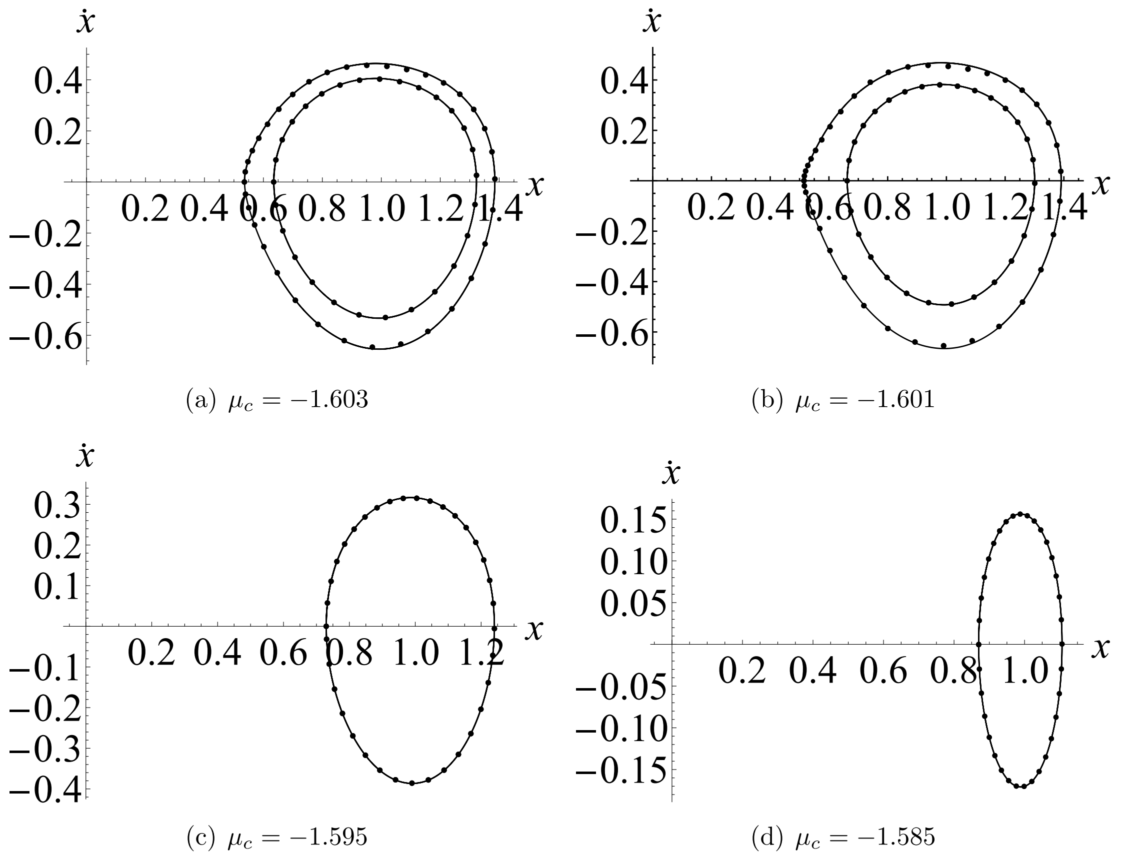


Figure 7. Limit cycles for oscillator (68) at different μ when $\varepsilon = 1$. '—' denote the results via the Runge-Kutta method. '···' denote the results via the present method.

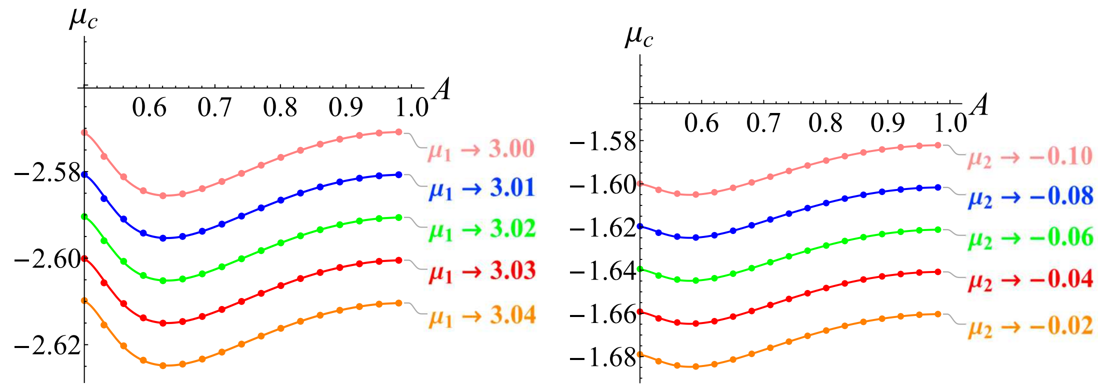
Table 5. The values of a_0, b, p_i and l_i for each μ_c in solutions (69)–(72).

μ_c	-1.603	-1.601	-1.595	-1.585
a_0	$-0.8500/-0.6886$	$-0.8755/-0.6391$	0.5107	-0.2364
b	1.3855/1.3235	1.3904/1.3010	1.2404	1.1063
p_0	0.6057/0.6684	0.5736/0.6746	0.6838	0.6909
p_2	$-0.1083/-0.0300$	$-0.1515/-0.0228$	-0.0124	-0.0038
p_4	$-0.0559/-0.0115$	$-0.0829/-0.0079$	-0.0033	-0.0004
p_6	$-0.0278/-0.0043$	$-0.0436/-0.0027$	-0.0009	-0.0001
p_8	$-0.0133/-0.0016$	$0.0220/-0.0009$	-0.0002	0
l_2	$-0.1146/-0.0951$	$-0.1149/-0.0879$	-0.0693	-0.0313
l_4	0.0123/0.0077	0.0140/0.0066	-0.0042	0.0009
l_6	$-0.0034/-0.0019$	$-0.0032/-0.0015$	-0.0008	-0.0001
l_8	$-0.0009/0$	$-0.0015/0$	0	0

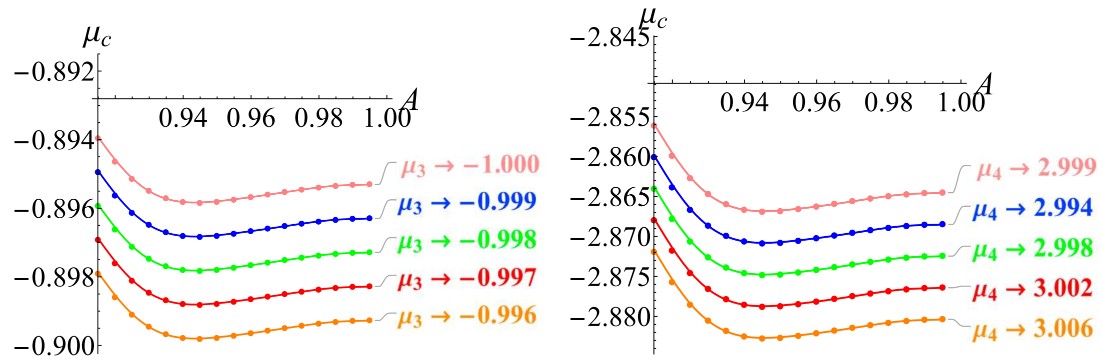
Table 6. Error analyses of the solutions in figure 7.

Figures	Amplitudes	MAEA[a]	RMSPE[b]	L_2-ND[c]
(a)	0.53540	0.0029546	0.05057421	1.695%
	0.63491	0.0053129	0.00484312	0.379%
(b)	0.51491	0.0010683	0.04347941	1.494%
	0.66142	0.0033677	0.00539951	0.270%
(c)	0.72970	0.0012832	0.00455645	0.128%
(d)	0.87010	1.1×10^{-6}	0.00099564	0.008%

[a] MAEA denotes the maximum absolute error in amplitude.
[b] RMSPE denotes the root mean square phase error over one period.
[c] L_2-ND denotes the relative L_2-norm differences.

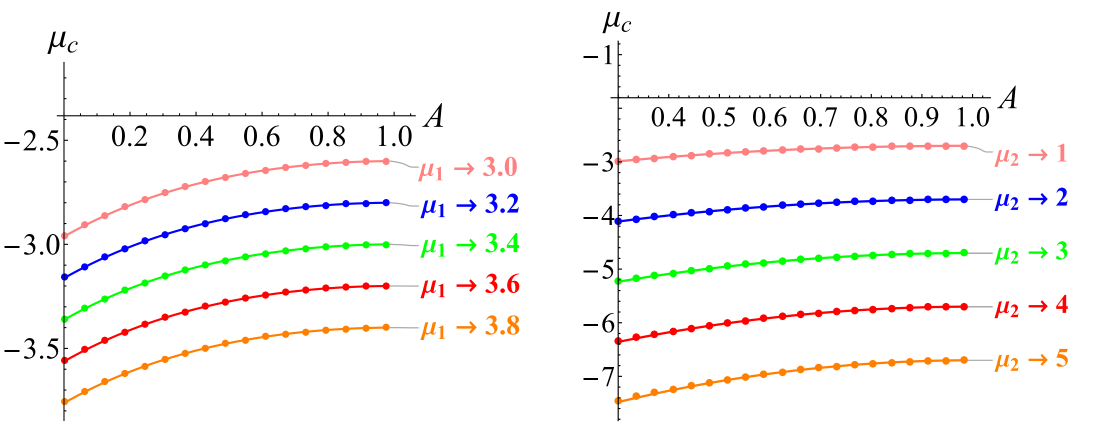


(a) $\alpha = 0.5,\ \beta = 0.1,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = -0.7$ (b) $\alpha = 0.5, \beta = 0.1, \mu_1 = 2, \mu_3 = -1, \mu_4 = -0.7$



(c) $\alpha = 0.9, \beta = 0.01, \mu_1 = 1, \mu_2 = -0.1, \mu_4 = 1$ (d) $\alpha = 0.9,\ \beta = 0.01,\ \mu_1 = 1,\ \mu_2 = -0.1,\ \mu_3 = -1$

Figure 8. The relationship between the amplitude A and control parameter μ_c of oscillator (68) when $\varepsilon = 0.1$. '—' denote the results via the present method. '···' denote the results via the Runge-Kutta method.

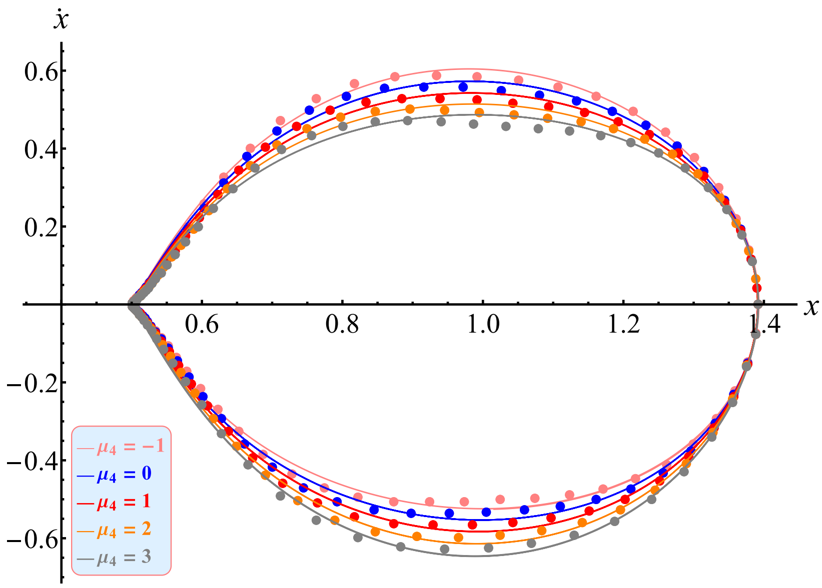


(a) $\alpha = 0,\ \beta = 0,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = 0.7$ (b) $\alpha = 0.3,\ \beta = 0,\ \mu_1 = 2,\ \mu_3 = -1,\ \mu_4 = 0.7$

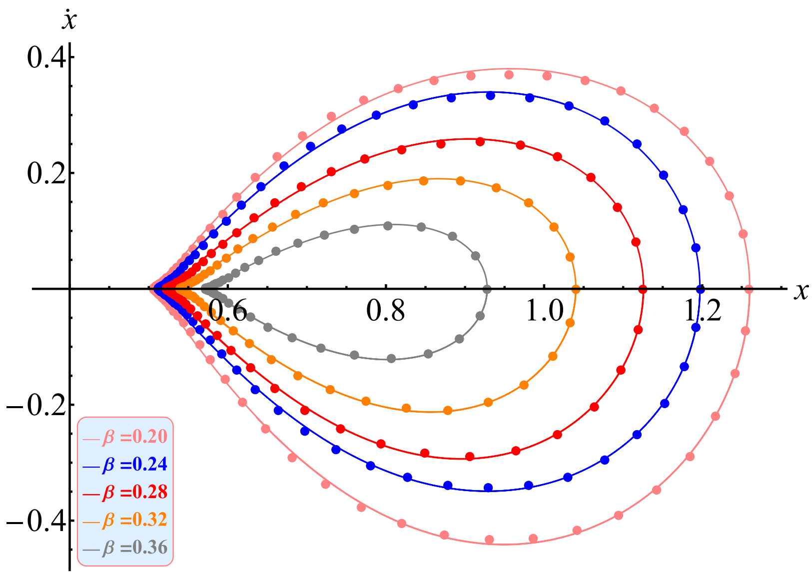
Figure 9. The relationship between the amplitude A and control parameter μ_c of oscillator (68) in its non-smooth case when $\varepsilon = 0.1$. '—' denote the results via the present method. '···' denote the results via the Runge-Kutta method.

(v) As the parameter μ_c continues to increases, Only one unstable small limit cycle persists and continues to shrink. Upon reaching $\mu_c = \mu_s^{(3)}$, this shrinking cycle eventually collapses into the singular point (0.9887, 0).

Furthermore, the presented method is capable of obtaining the approximate solution to oscillator (68). Considering the four cases of $\mu_c = -1.603$, $\mu_c = -1.601$, $\mu_c = -1.595$ and $\mu_c = -1.585$, the corresponding value of A can be obtained by using equations (56) and (59). From this, the first order approximate solution of



(a) $\alpha = 0.5,\ \beta = 0.2,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = 0.7$



(b) $\alpha = 0.5,\ \mu_1 = 2,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = 0.7$

Figure 10. The phase portraits of homoclinic solutions of oscillator (68) with different parameters when $\varepsilon = 0.5$. '—' denote the results via the Runge-Kutta method. '⋯' denote the results via the present method.

the small limit cycle can be expressed as:

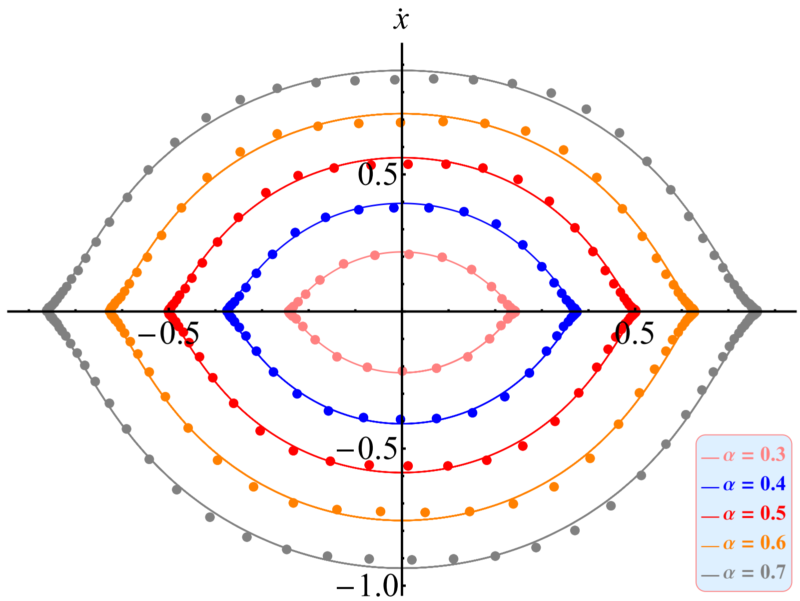
$$x = a_0 \cos^2 \varphi + b + O(\varepsilon^2) \tag{69}$$

$$\frac{d\varphi}{dt} = \Phi_0(\varphi) + \varepsilon\Phi_1(\varphi) + O(\varepsilon^2) \tag{70}$$

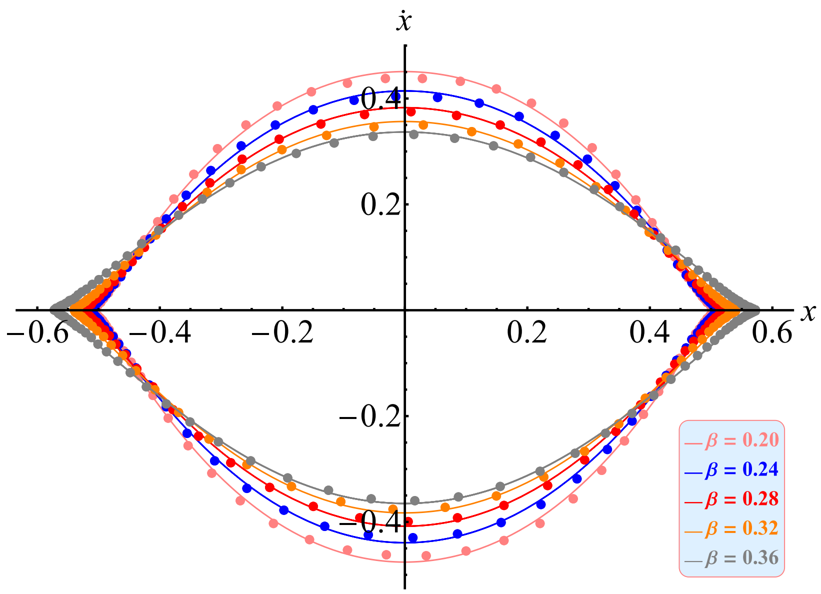
$$\Phi_0 = p_0 + p_2 \cos(2\varphi) + p_4 \cos(4\varphi) + p_6 \cos(6\varphi) + p_8 \cos(8\varphi) + HH \tag{71}$$

$$\Phi_1 = l_2 \sin(2\varphi) + l_4 \sin(4\varphi) + l_6 \sin(6\varphi) + l_8 sin(8\varphi) + HH \tag{72}$$

in which HH is used to denote the high order harmonics. The values of a_0, b, p_i and l_i for each μ_c are listed in table 5. When $\varepsilon = 1$, the phase portraits of the solution are shown in figure 7, and the comparison made in the figure is easy to find that the results obtained by this method are also accurate for moderately large ε. From the image of the solution, the result is consistent with that shown in figure 6, that is, for $\mu_s^{(0)} < \mu_c < \mu_s^{(2)}$, the larger small limit cycle evolves toward a homoclinic orbit as μ_c increases. Conversely, the smaller limit cycle shrinks, with its eventual convergence to (0.9887, 0) occurring when μ_c reaches $\mu_s^{(3)}$. To further display the accuracy of solutions depicted in this figure, the maximum absolute error in amplitude (MAEA), the root mean square phase error over one period (RMSPE) and relative L_2-norm differences (L_2-ND) are calculated and established in table 6. The indicators in this table all illustrate that the obtained solution has high accuracy.



(a) $\beta = 0.2,\ \mu_1 = 2,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = 0.7$



(b) $\alpha = 0.5,\ \mu_1 = 2,\ \mu_2 = -0.1,\ \mu_3 = -1,\ \mu_4 = 0.7$

Figure 11. The phase portraits of heteroclinic solutions of oscillator (68) with different parameters when $\varepsilon = 0.1$. '—' denote the results via the Runge-Kutta method. '···' denote the results via the present method.

In addition, as shown in figure 8, for the oscillator (68), when $\varepsilon = 0.1$, we consider 20 cases where there are two small limit cycle. And then substitute system parameters into the equation (56) to obtain the relationship between parameter μ_c and amplitude A, respectively. Similarly, with the aim of demonstrating the accuracy of the proposed method, the above curves are also drawn by numerical method and shown together in figure 8. Through comparison, it can be found that the present method has high precision. Particularly, when $\beta = 0$ and $0 < \alpha < 1$, equation (68) reduces to a non-smooth triple-well-like oscillator. To assess the accuracy of the present method under this condition, the $\mu_c - A$ curves for different α and μ_i are plotted in figure 9. A comparison with figure 8 shows that the if $|\varepsilon\mu_i|$ is small, the predictions of proposed method are remain reliable.

Next, as shown in figures 10 and 11, the homo-heteroclinic orbits of oscillator (68) are analyzed for different parameter sets. We consider only the right-hand side homoclinic orbit. From equation (44), we can get the solution of a_0 and b. Substituting them into equations (56) and (59), we obtain the homoclinic bifurcation parameter μ_{c-homo}. From equations (48) and (53), Φ_0 and Φ_1 can be solved. Then, we consider the heteroclinic orbit of oscillator (68). From equation (45), we can obtain the solution of a_0 and b. Substituting them into equations (56) and (58), we obtain the heteroclinic bifurcation parameter $\mu_{c-hetero}$. Φ_0 and Φ_1 can be obtained

Table 7. The values of a_0, b, p_i and l_i for each μ_1 in homoclinic solutions which shown in figure 10(a).

μ_1	−1	0	1	2	3
μ_c	1.31307266	0.34187456	−0.62932354	−1.60052164	−2.57171974
a_0	0.8908	0.8908	0.8908	0.8908	0.8908
b	0.5006	0.5006	0.5006	0.5006	0.5006
p_0	0.5338	0.5338	0.5338	0.5338	0.5338
p_2	0.2125	0.2125	0.2125	0.2125	0.2125
p_4	0.1284	0.1284	0.1284	0.1284	0.1284
p_6	0.0774	0.0774	0.0774	0.0774	0.0774
p_8	−0.0475	0.0774	0.0774	0.0774	0.0774
l_2	−0.0853	−0.0195	0.0463	0.1121	0.1778
l_4	0.0045	0.0091	0.0138	0.0184	0.0231
l_6	0.0065	0.0044	0.0023	0.0002	−0.0019
l_8	0.0019	0.0007	−0.0005	−0.0017	0.0030

Table 8. The values of a_0, b, p_i and l_i for each β in homoclinic solutions which shown in figure 10(b).

β	0.2	0.24	0.28	0.32	0.36
μ_c	−1.49842559	−1.45187451	−1.40129683	−1.34485352	−1.27698751
a_0	0.7534	0.6846	0.6026	0.4998	0.3560
b	0.5064	0.5125	0.5228	0.5402	0.5719
p_0	0.4456	0.4099	0.3715	0.3269	0.2667
p_2	0.2349	0.2328	0.2245	0.2082	0.1773
p_4	−0.1030	0.0894	0.0750	0.0598	0.0432
p_6	0.0468	0.0373	0.0294	0.0230	0.0174
p_8	−0.0242	−0.0192	−0.0157	−0.0130	−0.0105
l_2	0.0826	0.0718	0.0608	0.0488	0.0340
l_4	0.0094	0.0065	0.0041	0.0022	0.0009
l_6	0.0005	0.0004	0.0001	0.0001	0.0003
l_8	−0.0009	−0.0007	−0.0006	−0.0004	−0.0003

Table 9. The values of a_0, b, p_i and l_i for each α in homoclinic solutions which shown in figure 11(a).

α	0.3	0.4	0.5	0.6	0.7
μ_c	0.00115120	0.00191160	0.00148574	−0.00143239	−0.00888223
a_0	−0.4850	−0.7502	−1.0012	−1.2530	−1.5237
b	0.2425	0.3751	0.5006	0.6265	0.7619
p_0	0.3032	0.3679	0.3991	0.4111	0.4020
p_2	0.0067	0.0084	0.0089	0.0085	0.0066
p_4	0.1922	0.2231	0.2385	0.2506	0.2662
p_6	0.0062	0.0077	0.0082	0.0078	0.0061
p_8	−0.0573	−0.0770	−0.0888	−0.0951	−0.0938
l_2	−0.0616	−0.0953	−0.1249	−0.1501	−0.1690
l_4	−0.0010	−0.0013	−0.0010	0.0005	0.0043
l_6	0.0031	0.0063	0.0104	0.0152	0.0208
l_8	0	0.0001	0.0004	0.0009	0.0018

in the same way. From this, the first order approximate solution of the homo-heteroclinic orbit can be also expressed by equations (69)–(72). The values of a_0, b, p_i and l_i are listed in tables 7–10. To demonstrate their accuracy, the homo-heteroclinic orbits drawn by Runge-Kutta is also shown in figures 10 and 11. These are very close to values obtained by present method.

Finally, as shown in figures 12 and 13, with aid of the equation (56), we calculate the variation of homo-heteroclinic bifurcation parameter with different μ_i for oscillator (68). Similarly, with the aim of demonstrating the accuracy of the proposed method, when $\varepsilon = 0.1$, these curves are also drawn by numerical method and shown together in figures 12 and 13. Through comparison, it can be found that the present method has high precision and the predicted results are reliable. In particular, when $\beta = 0$ and $0 < \alpha < 1$, equation (68) reduces to a non-smooth triple-well-like oscillator. Its potential portrait resembles that shown in figure 5. Consequently, under this non-smooth condition, the oscillator exhibits a pair of homoclinic-like and heteroclinic-

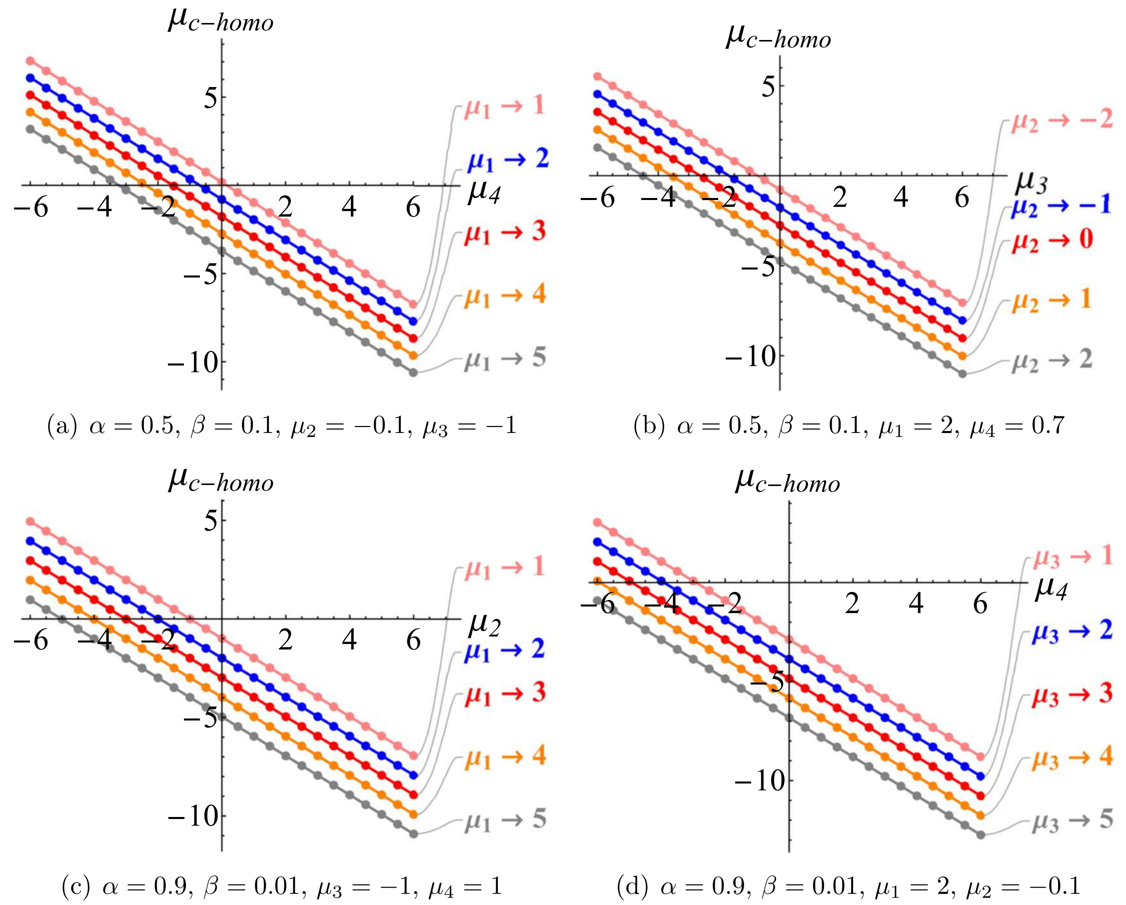


Figure 12. The variation of critical value of heteroclinic bifurcation parameter of oscillator (68) with system parameter $\mu_{c-hetero}$ when $\varepsilon = 0.1$. '—' denote the results via the Runge-Kutta method. '···' denote the results via the present method.

Table 10. The values of a_0, b, p_i and l_i for each β in homoclinic solutions which shown in figure 11(b).

β	0.2	0.24	0.28	0.32	0.36
μ_c	0.00142829	0.00136716	0.00125627	0.00104343	0.0005722
a_0	−1.0128	−1.0250	−1.0455	−1.0803	−1.1437
b	0.5064	0.5125	0.5228	0.5402	0.5719
p_0	0.2915	0.2590	0.2301	0.2034	0.1761
p_2	0.0055	0.0047	0.0040	0.0034	0.0027
p_4	0.2029	0.1866	0.1709	0.1561	0.1420
p_6	0.0051	0.0043	0.0037	0.0031	0.0025
p_8	−0.0485	−0.0387	−0.0306	−0.0235	−0.0159
l_2	−0.1175	−0.1166	−0.1166	−0.1177	−0.1201
l_4	−0.0008	−0.0008	−0.0007	−0.0005	−0.0001
l_6	0.0068	0.0060	0.0053	0.0048	0.0042
l_8	0.0002	0.0002	0.0002	0.0003	0.0004

like orbits. Moreover, we calculate how the corresponding homoclinic-like and heteroclinic-like bifurcation parameters vary with α. As shown in figure 14, the proposed method effectively predicts homoclinic-like and heteroclinic-like bifurcations.

6. Conclusions

The analytical study of the global evolution of limit cycles in coupled SD oscillators is hindered by the complexity of the irrational nonlinear potential, which exceeds the capability of many classical quantitative methods. To supplement this difficulty, the MGHFP method was adopted in this paper. Via this method, we can obtain the analytical relationship between the amplitude of the limit cycle and the control parameter μ_c, see

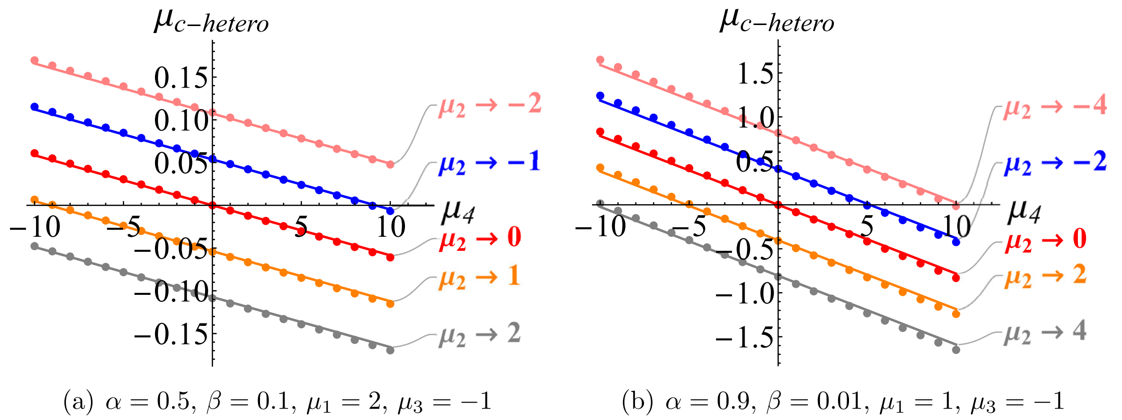


Figure 13. The variation of critical value of homoclinic-like and heteroclinic-like bifurcation parameters of oscillator (68) in its non-smooth case when $\varepsilon = 0.1$. '—' denote the results via the Runge-Kutta method. '⋯' denote the results via the present method.

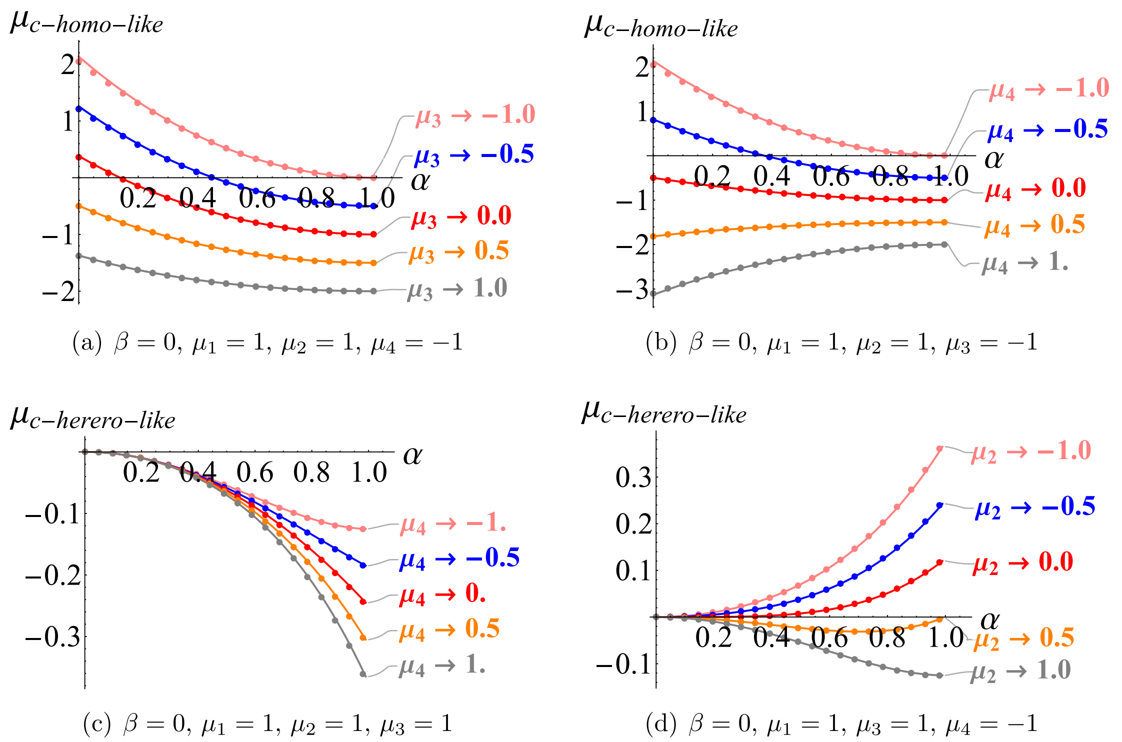


Figure 14. The variation of critical value of homoclinic-like and heteroclinic-like bifurcation parameters of oscillator (68) in its non-smooth case when e = 0.1. '—' denote the results via the Runge-Kutta method. '⋯' denote the results via the present method.

equation (56). In addition, the analytic criterion for the stability of the limit cycle is also obtained in equation (60). From the above results, the existence, stability, numbers and amplitude of limit cycles have been investigated quantitatively and analytically. Meanwhile, the critical values of homo-heteroclinic bifurcation parameter μ_c can be also calculated by equation (56). As the examples, we discuss the oscillator (46) in conditions of single well and triple well potential respectively in section 4. First, we have studied the large limit cycles for single well potential and the small limit cycles enclosed in homoclinic orbit for triple well potental. The relationship between the amplitude of the limit cycles and control parameter μ_c has been plotted for several sets of parameters. Compared with the numerical results, and the above figures are in good agreement. As the same time, we have selected a certain set of parameters and carried out detailed analysis and solution. The reliability of this method is further verified. In addition, we have also investigated the homo-heteroclinic orbits of the triple well system. We not only drew the figures of critical value of bifurcations parameters varying with the system parameters, but also calculated the homo-heteroclinic solutions with different parameters. The obtained results agree well with the numerical results, too. As demonstrated by the preceding examples, the proposed method enables a detailed analysis of the global evolution of limit cycles with respect to system parameters. This encompasses the prediction of limit cycle emergence, bifurcation behavior, convergence tendencies, and stability properties. Notably, the method proves particularly effective in accurately predicting the existence and evolution of unstable and semi-stable limit cycles, which are considerably challenging to locate via purely numerical means. Consequently, the framework presented in this work constitutes an effective analytical tool with both theoretical and practical significance.

Nevertheless, no method works universally. We would like to state that in the following conditions, accuracy of the present method will degrade.

(i) Large perturbation parameter ε. The method is fundamentally based on a first-order perturbation expansion, assuming ε is a small parameter. As shown in table 3, when $\varepsilon > 0.5$, the risk of the relative error exceeding 5% increases significantly. One should therefore restrict applications to $|\varepsilon\mu_i| \leqslant 0.5$ or consider extending the analysis to higher-order perturbations when larger nonlinearities are present.

(ii) Limit cycles crossing saddle points. As shown in Example 2, the method yields accurate results for homoclinic/heteroclinic orbits and the limit cycles they enclose in multi-stable oscillators. However, when a limit cycle crosses a saddle point, the oscillator's velocity drops sharply in the vicinity of the saddle, leading to abrupt features in the displacement waveform. In these cases, the Fourier series truncated at $m = 4$ may fail to capture the dynamics accurately, and a higher truncation order is required.

(iii) Non-smooth potential well. As illustrated in figure 4, the presence of non-smoothness can also affect the accuracy of the method. When studying such cases, one should either employ a smaller perturbation parameter ε or increase the truncation order.

Declarations

Funding

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Authors' contributions

Zhenbo Li: Writing–original draft, Methodology, Funding acquisition, Conceptualization. **Linxia Hou:** Writing–original draft, Writing-review & editing, Visualization. **Ruyue Peng**: Conceptualization, Writing-review & editing, Supervision.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

Appendix A. The expressions of P_{2i} and the values of their coefficients

$$\begin{aligned} p_0 = {} & 0.0131B_{10} + 0.6033B_{11} + 0.2095B_{12} + 0.1367B_{13} + 0.1115B_{14} + 0.1048B_{15} \\ & + 0.1115B_{16} + 0.1367B_{17} + 0.2095B_{18} + 0.6033B_{19} + 0.1531B_{20} + 0.0815B_{21} \\ & + 0.0605B_{22} + 0.0532B_{23} + 0.0532B_{24} + 0.0605B_{25} + 0.0815B_{26} + 0.1531B_{27} \\ & + 0.0131B_{28}, \end{aligned}$$

Table A1. The values of X_i, Y_i and Z_i in B_i.

i	X_i	Y_i	Z_i	i	X_i	Y_i	Z_i
11	0.9999	1.9848	0.0076	12	0.9955	1.8660	0.0670
13	0.9681	1.6428	0.1787	14	0.8918	1.3420	0.3290
15	0.7500	1.0000	0.5000	16	0.5500	0.6580	0.6710
17	0.3253	0.3572	0.8214	18	0.1295	0.1340	0.9330
19	0.0151	0.0152	0.9924	20	0.9991	1.9397	0.0302
21	0.9863	1.7660	0.1170	22	0.9375	1.5000	0.2500
23	0.8293	1.1736	0.4132	24	0.6556	0.8264	0.5868
25	0.4375	0.5000	0.7500	26	0.2203	0.2340	0.8830
27	0.0594	0.0603	0.9698				

$$\begin{aligned}
p_2 = {} & -0.0262B_{10} - 1.1882B_{11} - 0.3629B_{12} - 0.1758B_{13} - 0.0763B_{14} + 0.0763B_{16} \\
& + 0.1758B_{17} + 0.3629B_{18} + 1.1882B_{19} - 0.2878B_{20} - 0.1248B_{21} - 0.0605B_{22} \\
& - 0.0185B_{23} + 0.0185B_{24} + 0.0605B_{25} + 0.1248B_{26} + 0.2878B_{27} + 0.0262B_{28},
\end{aligned}$$

$$\begin{aligned}
p_4 = {} & 0.02612B_{10} + 1.1338B_{11} + 0.2095B_{12} - 0.0475B_{13} - 0.1708B_{14} - 0.2095B_{15} \\
& - 0.1708B_{16} - 0.0475B_{17} + 0.2095B_{18} + 1.1338B_{19} + 0.2346B_{20} + 0.0283B_{21} \\
& - 0.0605B_{22} - 0.1B_{23} - 0.1B_{24} - 0.0605B_{25} + 0.0283B_{26} + 0.2346B_{27} + 0.0262B_{28},
\end{aligned}$$

$$\begin{aligned}
p_6 = {} & -0.0262B_{10} - 1.0449B_{11} + 0.2369B_{13} + 0.1931B_{14} - 0.1931B_{16} - 0.2369B_{17} \\
& + 1.0449B_{19} - 0.1531B_{20} + 0.0815B_{21} + 0.121B_{22} + 0.0532B_{23} - 0.0532B_{24} \\
& - 0.121B_{25} - 0.0815B_{26} + 0.1531B_{27} + 0.0262B_{28},
\end{aligned}$$

$$\begin{aligned}
p_8 = {} & 0.0262B_{10} + 0.9243B_{11} - 0.2095B_{12} - 0.257B_{13} + 0.0387B_{14} + 0.2095B_{15} \\
& + 0.0387B_{16} - 0.257B_{17} - 0.2095B_{18} + 0.9243B_{19} + 0.0532B_{20} - 0.1531B_{21} \\
& - 0.0605B_{22} + 0.0815B_{23} + 0.0815B_{24} - 0.0605B_{25} - 0.1531B_{26} + 0.0532B_{27} \\
& + 0.0262B_{28},
\end{aligned}$$

where

$$B_{10} = \sqrt{-\frac{1}{a_0}\left(\left(\alpha + b\right)\left(1 - \frac{1}{\sqrt{\beta^2 + (-\alpha + b)^2}}\right) + (\alpha + b)(1 - \frac{1}{\sqrt{\beta^2 + (\alpha + b)^2}}\right)\right)},$$

$$B_{28} = \left(-\frac{1}{a_0}\left((-\alpha + a_0 + b)\left(1 - \frac{1}{\sqrt{\beta^2 + (-\alpha + a_0 + b)^2}}\right) + (\alpha + a_0 + b)\left(1 - \frac{1}{\sqrt{\beta^2 + (\alpha + a_0 + b)^2}}\right)\right)\right)^{\frac{1}{2}},$$

$$\begin{aligned}
B_i = {} & \left(\frac{1}{a_0^2}(X_i a_0^2 + Y_i a_0 b + \sqrt{\beta^2 + (-\alpha + Z_i a_0 + b)^2} + \sqrt{\beta^2 + (\alpha + Z_i a_0 + b)^2}\right. \\
& \left. - \sqrt{\beta^2 + (-\alpha + a_0 + b)^2} - \sqrt{\beta^2 + (\alpha + a_0 + b)^2})\right)^{\frac{1}{2}}, (i = 11, 12, \cdots, 27).
\end{aligned}$$

The values of X_i, Y_i and Z_i are listed in table A1.

Appendix B. The expressions of C

$$\begin{aligned}
C = {} & \left(1 - C_0 + \frac{\sqrt{24(-1 + C_0)^2 + 8C_1 + \frac{32 2^{1/3} C_2}{C_5} + 2^{2/3} C_5}}{2\sqrt{6}} + \frac{1}{2}(8(-1 + C_0)^2\right. \\
& \left. + \frac{8C_1}{3} - \frac{16 2^{1/3} C_2}{3C_5} - \frac{C_5}{3 2^{1/3}} + \frac{8\sqrt{6}(-2(-1 + C_0)^3 - (-1 + C_0)C_1 + C_3)}{\sqrt{24(-1 + C_0)^2 + 8C_1 + \frac{32 2^{1/3} C_2}{C_5} + 2^{2/3} C_5}}\right)^{\frac{1}{2}}\Bigg)^{\frac{1}{2}},
\end{aligned}$$

where A denotes the amplitude and

$$
\begin{aligned}
C_0 &= -A^2 + \sqrt{A^2 - 2A\alpha + \alpha^2 + \beta^2} + \sqrt{A^2 + 2A\alpha + \alpha^2 + \beta^2},\\
C_1 &= 2\alpha^2 + 2\beta^2 + 4C_0 - 3C_0^2,\\
C_2 &= 12\alpha^2 + \alpha^4 + 2\alpha^2\beta^2 + \beta^4 - 14\alpha^2 C_0 - 2\beta^2 C_0 + C_0^2,\\
C_3 &= -4\alpha^2 + 2\alpha^2 C_0 + 2\beta^2 C_0 + C_0^2 - C_0^3,\\
C_4 &= -4\alpha^2 C_0^2 - 4\beta^2 C_0^2 + C_0^4,\\
C_5 &= (-16C_1^3 - 288(C_0 - 1)C_1C_3 + 432C_3^2 + 432(C_0 - 1)^2 C_4 + 144C_1C_4\\
&\quad + (-4(16\alpha^4 + 32\alpha^2\beta^2 + 192\alpha^2 + 16\beta^4 - 224\alpha^2 C_0 - 32\beta^2 C_0 + 16C_0^2)^3\\
&\quad + (-16C_1^3 - 288(C_0 - 1)C_3C_1 + 144C_4C_1 + 432C_3^2 + 432(C_0 - 1)^2 C_4)^2)^{\frac{1}{2}})^{\frac{1}{3}}.
\end{aligned}
$$

References

[1] Chen C-T 1984 *Linear System Theory and Design* (Saunders College Publishing)

[2] Lu K and Yang Q 2026 N heteroclinic cycles coexisting and chaos in a 3d (n+1)-zone piecewise smooth system *Int. J. Bifurcation Chaos* **36** 26500562026

[3] Li Y, Wei Z and Moroz I 2025 Melnikov-type method and homoclinic bifurcation in a class of hybrid piecewise smooth systems under noise and impulsive excitation *Journal of Nonlinear Science* **35** 59

[4] Mehmood T, Rafaqat M, Saleem S and Merga F E 2025 Machine learning and bifurcation analysis in a discrete predator-prey model with neem-induced mortality *Sci. Rep.* **15** 40792

[5] Hu S and Zhou L 2025 Stochastic hopf bifurcation and random chaos of a multi-stable rotational energy harvesting system *Chaos Solitons & Fractals* **199** 116850

[6] Li Y, Wei Z, Zhang W and Kapitaniak T 2023 Melnikov-type method for chaos in a class of hybrid piecewise-smooth systems with impact and noise excitation under unilateral rigid constraint *Appl. Math. Modelling* **122** 506–23

[7] Cao Q, Wiercigroch M, Pavlovskaia E E, Grebogi C and T. Thompson J M 2006 Archetypal oscillator for smooth and discontinuous dynamics *Phys. Rev. E* **74** 046218

[8] Cao Q, Wiercigroch M, Pavlovskaia E, Grebogi C, Michael J and Thompson T 2008 The sd oscillator and its attractors *J. Phys. Conf. Ser.* **96** 012064

[9] Li Z-X, Cao Q-J, Wiercigroch M and Leger A 2013 Analysis of the periodic solutions of a smooth and discontinuous oscillator *Acta Mech. Sin.* **29** 575–82

[10] Chen H and Xie J 2016 Harmonic and subharmonic solutions of the sd oscillator *Nonlinear Dyn.* **84** 2477–86

[11] Liang Z and Yang Y 2020 Existence and stability of periodic oscillations of a smooth and discontinuous oscillator *Physica A-Statistical Mechanics and Its Applications* **555** 124511

[12] Santhosh B, Narayanan S and Padmanabhan C 2016 Discontinuity induced bifurcations in nonlinear systems *Procedia IUTAM* **19** 219–27

[13] Hao Z, Cao Q and Wiercigroch M 2017 Nonlinear dynamics of the quasi-zero-stiffness sd oscillator based upon the local and global bifurcation analyses *Nonlinear Dyn.* **87** 987–1014

[14] Yi T and Du Z 2014 Degenerate grazing bifurcations in a simple bilinear oscillator *Int. J. Bifurcation Chaos* **24** 1450141

[15] Tian R, Wu Q, Yang X and Si C 2013 Chaotic threshold for the smooth-and-discontinuous oscillator under constant excitations *European Physical Journal Plus* **128** 80

[16] Yang D and Zhou J 2014 Connections among several chaos feedback control approaches and chaotic vibration control of mechanical systems *Commun. Nonlinear Sci. Numer. Simul.* **19** 3954–68

[17] Li S, Ma X, Bian X, Lai S-K and Zhang W 2020 Suppressing homoclinic chaos for a weak periodically excited non-smooth oscillator *Nonlinear Dyn.* **99** 1621–42

[18] Peng R, Li Q and Zhang W 2024 Chaos analysis of sd oscillator with two-frequency excitation *Nonlinear Dyn.* **112** 7649–77

[19] Peng R, Li Q and Zhang W 2024 Homoclinic bifurcation analysis of a class of conveyor belt systems with dry friction and impact *Chaos Solitons & Fractals* **180** 114469

[20] Chen H, Llibre J and Tang Y 2018 Global dynamics of a sd oscillator *Nonlinear Dyn.* **91** 1755–77

[21] Li Z, Cao Q and Nie Z 2020 Stick-slip vibrations of a self-excited sd oscillator with coulomb friction *Nonlinear Dyn.* **102** 1419–35

[22] Li Z, Han N and Yang T 2024 Friction-induced vibrations of a two dof self-excited sd oscillator *Journal of Vibration Engineering & Technologies* **12** 601–18

[23] Tian R, Cao Q and Yang S 2010 The codimension-two bifurcation for the recent proposed sd oscillator *Nonlinear Dyn.* **59** 19–27

[24] YanWei H, QingJie C, YuShu C and Wiercigroch M 2012 A novel smooth and discontinuous oscillator with strong irrational nonlinearities *Science China-Physics Mechanics & Astronomy* **55** 1832–43

[25] Cao Q J, Han Y W, Liang T W, Wiercigroch M and Piskarev S 2014 Multiple buckling and codimension-three bifurcation phenomena of a nonlinear oscillator *Int. J. Bifurcation Chaos* **24** 1430005

[26] Lai S K, Wu B S and Lee Y Y 2017 Free vibration analysis of a structural system with a pair of irrational nonlinearities *Appl. Math. Modelling* **45** 997–1007

[27] Zhou B, Jin Y and Xu H 2022 Subharmonic resonance and chaos for a class of vibration isolation system with two pairs of oblique springs *Appl. Math. Modelling* **108** 427–44

[28] Zhou B, Jin Y and Xu H 2022 Global dynamics for a class of tristable system with negative stiffness *Chaos Solitons & Fractals* **162** 112509

[29] Zhou B, Jin Y and Xu H 2022 Homoclinic-heteroclinic bifurcations and chaos in a coupled sd oscillator subjected to gaussian colored noise *Int. J. Bifurcation Chaos* **32** 2250240

[30] Han Y, Cao Q, Chen Y and Wiercigroch M 2015 Chaotic thresholds for the piecewise linear discontinuous system with multiple well potentials *Int. J. Non-Linear Mech.* **70** 145–52

[31] Chen H and Tang Y 2020 An oscillator with two discontinuous lines and van der pol damping *Bulletin des Sciences Mathematiques* **161** 102867
[32] Azizi S, Rezazadeh G, Ghazavi M-R and Khadem S E 2011 Stabilizing the pull-in instability of an electro-statically actuated micro-beam using piezoelectric actuation *Appl. Math. Modelling* **35** 4796–815
[33] Azizi S, Haddad Khodaparast H, Madinei H, Younis M and der Rezazadeh I 2025 Bifurcation-based dynamics and internal resonance in micro ring resonators for mems applications *Nonlinear Dyn.* **113** 24329–42
[34] Azizi S, Ghazavi M-R, Rezazadeh G and Khadem S E 2015 Thermo-elastic damping in a functionally graded piezoelectric micro-resonator *Int. J. Mech. Mater. Des.* **11** 357–69
[35] Soltani K, Rezazadeh G and Henry M P 2023 Nonlinear dynamics of a broadband vortex-induced vibration–based energy harvester *J. Eng. Mech.* **149** 04023046
[36] Chen S H and Cheung Y K 1997 An elliptic lindstedt-poincare method for certain strongly non-linear oscillators *Nonlinear Dyn.* **12** 199–213
[37] Chen S H, Yang X M and Cheung Y K 1998 Periodic solutions of strongly quadratic non-linear oscillators by the elliptic perturbation method *J. Sound Vib.* **212** 771–80
[38] Li Z, Tang J and Cai P 2013 A generalized harmonic function perturbation method for determining limit cycles and homoclinic orbits of helmholtz-duffing oscillator *J. Sound Vib.* **332** 5508–22
[39] Li Z, Tang J and Cai P 2016 Predicting homoclinic and heteroclinic bifurcation of generalized duffing-harmonic-van de pol oscillator *Qualitative Theory of Dynamical Systems* **15** 19–37
[40] Li Z, Hou L, Zhang Y and Xu F 2024 A modified perturbation method for global dynamic analysis of generalized mixed rayleigh-liénard oscillator with cubic and quintic nonlinearities *Phys. Scr.* **99** 075213
[41] Li Z, Cai J and Hou L 2024 A modified generalized harmonic function perturbation method and its application in analyzing generalized duffing-harmonic-rayleigh-liénard oscillator *Int. J. Non-Linear Mech.* **166** 104832
[42] Li Z, Hou L and Peng R 2025 Global evolution of limit cycles and homoclinic bifurcation of smooth and discontinuous oscillator with quartic nonlinear damping *Int. J. Non-Linear Mech.* **178** 105185
[43] Yuan J, Jin G, Ye T, Sun Z and Bai J 2025 Incorporating electromagnetic transducers in quasi-zero stiffness isolators for nonlinearity mitigation and energy harvesting *Nonlinear Dyn.* **113** 16031–82
[44] Yu N, Tian F, Ma H, Cai C, Tang Y and Yan B 2025 Tunable quasi-zero stiffness vibration isolators with electromagnetic damping for low-frequency vibration isolation enhancement *Mech. Syst. Signal Process.* **237** 113082
[45] Liu Z, Xu Y, Lv Q, Wang X, Chen Y, Dai H, Wei X and Huan R 2024 Amplitude deflection in a nonlinear mems resonator under parametric excitation *Int. J. Non-Linear Mech.* **163** 104754
[46] Lu S, Yu X and An Z 2025 Bifurcation and dynamics of periodic solutions of mems model with squeeze film damping *Nonlinear Analysis-Real World Applications* **81** 104229
[47] Davis P J and Rabinowitz P 2007 *Methods of Numerical Integration* (Courier Corporation)
[48] Dumortier F, Llibre J and Artés J C 2006 *Qualitative Theory of Planar Differential Systems* (Springer)